

Synaptic enrichment and dynamic regulation of the two opposing dopamine receptors within the same neurons

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eLife Assessment

This study uses state-of-the-art methods to label endogenous dopamine receptors in a subset of *Drosophila* mushroom body neuronal types. The authors report that Dop1R1 and Dop2R receptors, which have opposing effects on intracellular cAMP, are present in axons termini of Kenyon cells, as well as those of two classes of dopaminergic neurons that innervate the mushroom body indicative of autocrine modulation by dopaminergic neurons. Additional experiments showing opposing effects of starvation on Dop1R1 and Dop2R levels in mushroom body neurons are consistent with a role for dopamine receptor levels increasing the efficiency of learned food-odour associations in starved flies. Supported by **solid** data, this is an **important** contribution to the field.

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Abstract

Dopamine can play opposing physiological roles depending on the receptor subtype. In the fruit fly *Drosophila melanogaster*, *Dop1R1* and *Dop2R* encode the D₁- and D₂-like receptors, respectively, and are reported to oppositely regulate intracellular cAMP levels. Here, we profiled the expression and subcellular localization of endogenous Dop1R1 and Dop2R in specific cell types in the mushroom body circuit. For cell-type-specific visualization of endogenous proteins, we employed reconstitution of split-GFP tagged to the receptor proteins. We detected dopamine receptors at both presynaptic and postsynaptic sites in multiple cell types. Quantitative analysis revealed enrichment of both receptors at the presynaptic sites, with Dop2R showing a greater degree of localization than Dop1R1. The presynaptic localization of Dop1R1 and Dop2R in dopamine neurons suggests dual feedback regulation as autoreceptors. Furthermore, we discovered a starvation-dependent, bidirectional modulation of the presynaptic receptor expression in the PAM and PPL1

clusters, two distinct subsets of dopamine neurons, suggesting regulation of appetitive behaviors. Our results highlight the significance of the co-expression of the two opposing dopamine receptors in the spatial and conditional regulation of dopamine responses in neurons.

Introduction

Neurotransmitters typically have multiple cognate receptors, and they may recruit different second messenger systems. Therefore, the expression and localization of receptor subtypes are critical for determining cellular responses to neurotransmitter inputs. The dopaminergic system offers an ideal *in-vivo* study case to this end, as it regulates a wide array of physiological functions through combinations of different receptor subtypes. In mammals, D₁-like receptors are coupled with G_{αs}, thereby activating adenylate cyclase upon ligand binding, whereas G_{αi}-coupled D₂-like receptors inhibit cyclase activity (Beaulieu & Gainetdinov, 2011). Four dopamine receptors have been identified in *Drosophila*: *Dop1R1*, *Dop1R2*, *Dop2R*, and *DopEcR* (K. A. Han et al., 1996; Hearn et al., 2002; Srivastava et al., 2005; Sugamori et al., 1995). *Dop1R1* and *Dop2R*, also known as *DopR1*, *dDA1* and *dumb*, and as *DD2R*, respectively, correspond to the D₁- and D₂-like receptors, respectively (Hearn et al., 2002; Sugamori et al., 1995). *Dop1R2* and *DopEcR* are invertebrate-specific and have been reported to recruit different second messenger systems (K. A. Han et al., 1996; Srivastava et al., 2005). Intriguingly, recent data of single-cell RNA-seq and transgenic expression profiling revealed that the expression of these dopamine receptors is highly overlapping in the fly brain (Croset et al., 2018; Davie et al., 2018; Kondo et al., 2020), unlike the spatially segregated expression of the D₁- and D₂-like receptors in vertebrate brains (Gerfen & Surmeier, 2011). Considering the opposing physiological roles of the *Dop1R1* and *Dop2R*, their protein localization, especially in those cells where they are co-expressed, should be critical to determine the responses to the dopamine inputs.

Drosophila mushroom bodies (MB) have long served as a unique dopaminergic circuit model to study adaptive behaviors, such as associative learning. MB-projecting neurons and their connections have been systematically described at both mesoscopic and ultrastructural resolutions (Aso, Hattori, et al., 2014; Li et al., 2020; Takemura et al., 2017; Tanaka et al., 2008). Kenyon cells (KCs), the major MB intrinsic neurons, encode a variety of sensory information (Honegger et al., 2011; Turner et al., 2008; Vogt et al., 2014). Intriguingly, each KC receives synaptic inputs from different dopaminergic projections in multiple spatially segmented compartments along its axon in the MB lobe (Aso, Hattori, et al., 2014; Tanaka et al., 2008). MB-projecting dopamine neurons (DANs) originate from the three cell clusters (Mao & Davis, 2009; Nässel & Elekes, 1992). DANs in the protocerebral posterior lateral 1 (PPL1) cluster project to the vertical lobes and the peduncle of the MB, and they control different aspects of associative memory (Aso et al., 2010, 2012; Aso & Rubin, 2016; Claridge-Chang et al., 2009; Krashes et al., 2009; Mao & Davis, 2009; Masek et al., 2015; Riemensperger et al., 2005; Takemura et al., 2017; Vogt et al., 2014). The protocerebral anterior medial (PAM) cluster, the largest DAN cluster, mostly projects to the medial lobes of the MB, and many PAM neurons are involved in reward processing (Burke et al., 2012; Felsenberg et al., 2018; Huetteroth et al., 2015; Ichinose et al., 2021; Lin et al., 2014; Liu et al., 2012; Yamagata et al., 2015, 2016). DANs in PPL2ab project to the MB calyx and control the conditioned odor response (Boto et al., 2019). In addition to the variety of the dopamine sources, KCs express all four dopamine receptor subtypes (Deng et al., 2019; Kondo et al., 2020). Besides KCs, these lobe-projecting DANs have synaptic outputs to multiple types of neurons, including MBONs, APL and DPM (Li et al., 2020; Takemura et al., 2017; Zhou et al., 2019). Given the multitude of modulatory effects of dopamine in the MB (Berry et al., 2018; Cohn et al., 2015; Handler et al., 2019), receptor localization in each cell type provides important information for interpreting such functional diversity.

The projections and synapses of the neurons in the MB circuit are tightly intertwined (Li et al., 2020 [↗](#); Takemura et al., 2017 [↗](#)). Therefore, conventional immunohistochemical approaches using light microscopy do not allow identification of cells from which immunoreactive signals originate. Precise determination of their subcellular localization requires conditional visualization of the proteins of interest only in the target MB neurons (Bonanno et al., 2024 [↗](#); Fendl et al., 2020 [↗](#); Sanfilippo et al., 2024 [↗](#)). Employing the CRISPR/Cas9-mediated split-GFP tagging strategy (D. Kamiyama et al., 2016 [↗](#); R. Kamiyama et al., 2021 [↗](#); Kondo et al., 2020 [↗](#)), we profiled the spatial distribution of endogenous Dop1R1 and Dop2R proteins in KCs, the PAM, and the PPL1 DANs.

Results

Co-expression of *Dop1R1* and *Dop2R* genes in the adult *Drosophila* brain

To compare the expression of different dopamine receptor genes in detail, we used T2A-GAL4 knock-ins of the endogenous *Dop1R1* and *Dop2R* genes (Kondo et al., 2020 [↗](#)) with fluorescent reporters. Both lines labelled many neuropils including the MB (Figure 1A [↗](#) and 1B [↗](#)), and the overlapping expression of these receptors is consistent with our previous quantification of GAL4-positive cells for both genes (58,049 and 68,528 for *Dop1R1* and *Dop2R*, respectively, out of 118,331 brain cells; Kondo et al., 2020 [↗](#)). Double labelling of *Dop1R1-T2A-LexA* and *Dop2R-T2A-GAL4* expression revealed cells with overlapping and differential patterns (Figure 1C [↗](#)–1I [↗](#); see also Kondo et al., 2020 [↗](#)). We confirmed the co-expression of *Dop1R1* and *Dop2R* in the KCs (Figure 1 – figure supplement 1 [↗](#)) as reported previously (Kondo et al., 2020 [↗](#)). The PAM cluster of DANs expressed both *Dop1R1* and *Dop2R* (Figure 1C [↗](#) and 1D [↗](#)). On the other hand, most of the DANs in the PPL1 cluster strongly expressed *Dop2R*, but *Dop1R1* only weakly (Figure 1C [↗](#) and 1E [↗](#)). We further found that *Dop1R1*, but not *Dop2R*, was highly expressed in the ring neurons projecting to the ellipsoid body (Figure 1G [↗](#) and 1H [↗](#); Hanesch et al., 1989 [↗](#)) and the neuropil ensheathing glia (Figure 1I [↗](#); Awasaki et al., 2008 [↗](#)). In conclusion, *Dop1R1* and *Dop2R* genes are co-expressed in PAM neurons and KCs but have selective expressions in other cell types.

Quantification of subcellular enrichment of endogenous proteins in target cells

The overlapping expression of *Dop1R1* and *Dop2R* genes prompted us to examine the subcellular localization of these receptor proteins. To elucidate the localization of these broadly expressed receptors (Figure 1 [↗](#)), we took advantage of split-GFP tagging of endogenous proteins (D. Kamiyama et al., 2016 [↗](#); Kondo et al., 2020 [↗](#)). By adding seven copies of GFP₁₁ tags to the C-termini of the *Dop1R1* and *Dop2R* proteins, their intracellular distribution can be visualized specifically in cells expressing GFP₁₋₁₀ through split-GFP reconstitution (Figure 2A [↗](#)). To verify the functional integrity of reconstituted GFP (rGFP)-fused receptors, we examined aversive olfactory memory of homozygous flies carrying GFP₁₁-tagged dopamine receptors and induced GFP reconstitution in KCs. Both *Dop1R1* and *Dop2R* have been shown to be required for aversive memory in α/β and γ KCs (Kim et al., 2007 [↗](#); Scholz-Kornehl & Schwärzel, 2016 [↗](#)). Memory scores of the rGFP-fused receptor flies were comparable to that of control flies without GFP₁₁ insertion (Figure 2 – figure supplement 1 [↗](#)).

To quantify the subcellular enrichment of the receptors, we devised Localization Index (LI). Briefly, LI is the normalized ratio of the rGFP signal to the reference membrane marker signal (CD4::tdTomato). If the target and reference proteins had the identical distribution, LI would be 1 everywhere. More or less enrichment of rGFP would result in the LI larger or smaller than 1, respectively (Figure 2C [↗](#); see also Methods for detail). For visualization, the reference membrane

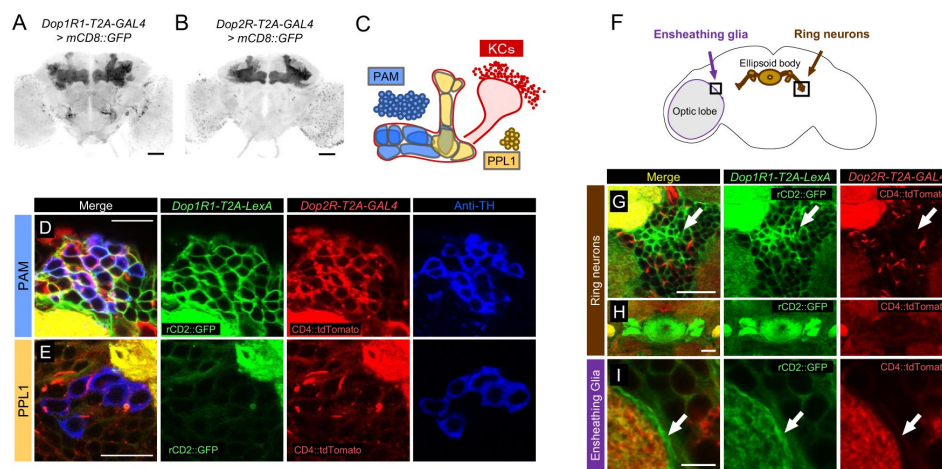


Figure 1.

Co-expression of *Dop1R1* and *Dop2R* genes in adult *Drosophila* brain.

(A and B) The expression of *Dop1R1-T2A-GAL4* and *Dop2R-T2A-GAL4* visualized by *UAS-mCD8::GFP*. Maximum-intensity projections of the whole brain.

(C) Schematic of the KCs and the MB-innervating dopamine neurons from the PAM and PPL1 clusters.

(D-E, G-I) Double labelling of *Dop1R1-T2A-LexA* and *Dop2R-T2A-GAL4* expressions by *lexAop-rCD2::GFP* (green) and *UAS-CD4::tdTomato* (red), respectively. Dopamine neurons were immunostained with anti-TH antibody (blue). Single optical sections are shown. Cell bodies of the PAM cluster (D), the PPL1 cluster (E), ring neurons projecting to the ellipsoid body (G and H) and ensheathing glia (I) are shown.

(F) Schematic of the regions shown in (G-I).

Scale bars, 50 μ m (A and B), 5 μ m (D, E and I), 20 μ m (G and H).

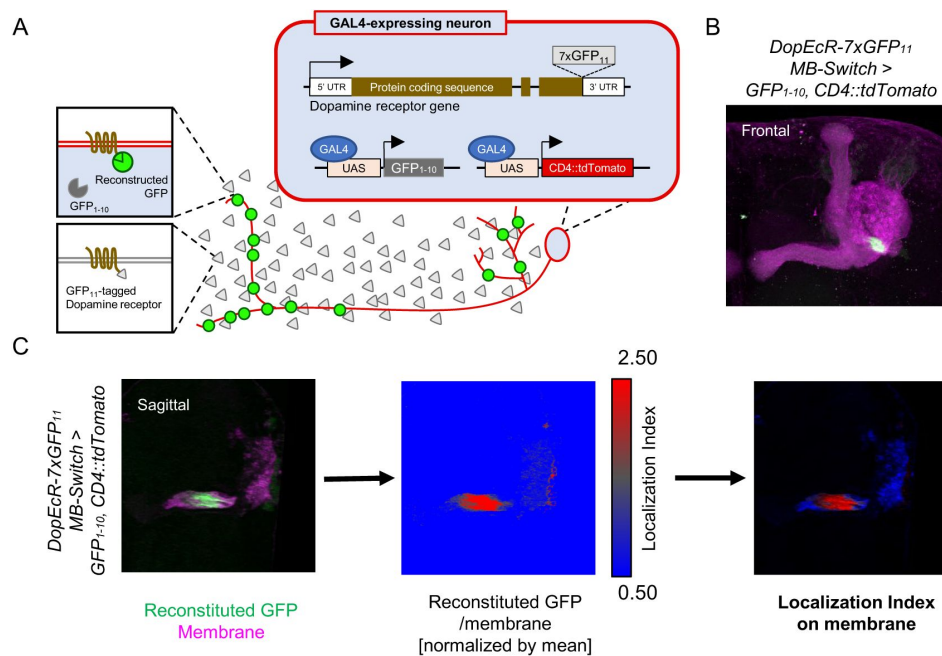


Figure 2.

Cell type-specific visualization of endogenous proteins with GFP₁₁ tag.

(A) Principle of cell type specific fluorescent labelling of target proteins by GFP₁₁ tag. Seven copies of GFP₁₁ are fused to the C-terminal of endogenous receptors. GFP₁₋₁₀ and membrane marker CD4::tdTomato are expressed in the target cells by GAL4/UAS system. In the target cells, reconstitution of GFP occurs on the endogenous proteins tagged with GFP₁₁.

(B) As an example, DopEcR::GFP₁₁ is visualized in KCs using *MB-Switch*, a ligand-inducible GAL4 driver. To activate Gene-Switch, flies were fed with food containing 200 μ M RU486 for 12 hours before dissection. A merged image of reconstituted GFP (green) and cellular membrane visualized by CD4::tdTomato (magenta). Maximum intensity projection of the whole left MB.

(C) The workflow for visualizing subcellular protein enrichment by localization index (LI). A single sagittal section of the MB calyx and peduncle is shown. The ratio of reconstituted GFP to membrane signal is calculated and normalized by the mean of all voxels to provide LI. In the middle image, LI is color-coded so that red represents local receptor enrichment. In the right image, the intensity of LI color is adjusted based on the membrane signal.

signal was color-coded with LI (**Figure 2C**). As proof of principle, mapping the LI of DopEcR::rGFP signals in KCs highlighted enrichment in the proximal peduncle (**Figure 2B** and **2C**), which is consistent with the previous report (Kondo et al., 2020). This representation thus visualizes the subcellular enrichment of the targeted receptors in the plasma membrane of GAL4-expressing cells.

Colocalization of Dop1R1 and Dop2R proteins

First, we compared the localization of Dop1R1 and Dop2R proteins in KCs, where both receptor genes were highly expressed (**Figure 1 – figure supplement 1**). These receptors were distributed throughout KC membranes with predominant enrichment in the lobes (**Figure 3A–3C**), but sparsely in the calyx (**Figure 3B** and **3C**). LI quantification revealed that enrichment in the lobes was more pronounced in Dop2R compared to Dop1R1, whereas localization to the calyx was sparser in Dop2R than Dop1R1 (**Figure 3D**). We confirmed that the differential localization of these receptors was consistent across multiple experimental batches conducted on different days (**Figure 3 – figure supplement 1**).

KCs have major presynaptic sites in the MB lobes, where Dop1R1 and Dop2R are enriched (**Figure 3A–3D**). Therefore, we examined receptor localization with the reference of Brp immunostaining, which labels the active zones (AZ) (Wagh et al., 2006). To distinguish the AZ in KCs from those in the other neurons, we co-labeled the plasma membrane of KCs and conducted high-magnification imaging using Airyscan. At this resolution, Brp puncta in KCs and those from non-KCs could be distinguished based on their overlap with the KC membrane marker (**Figure 4B** and **4C**). Interestingly, KC-specific visualization of Dop1R1 and Dop2R proteins revealed signals around the Brp puncta of the same cells, suggesting presynaptic localization (**Figure 4B** and **4C**). Additionally, we found the receptor condensates adjacent to the Brp clusters of non-KCs, suggesting their localization at the postsynaptic sites (**Figure 4B** and **4C**). The synaptic receptor localization is consistent in different subsets of KCs (α/β and γ ; **Figure 4A–4C**). In conclusion, both of these antagonizing dopamine receptors are enriched in the presynaptic and postsynaptic sites of KCs (**Figure 4D**).

To further clarify the presynaptic localization in KCs, we labeled the AZ of KCs by expressing *Brp^{short}::mStraw* (Fouquet et al., 2009) and confirmed that both Dop1R1::rGFP and Dop2R::rGFP were associated with the Brp puncta in the lobes (**Figure 5A**, **Figure 5 – figure supplement 1**). Interestingly, we found that not all Brp puncta of KCs were associated with the dopamine receptors (**Figure 5A**), suggesting that dopaminergic presynaptic modulation is heterogeneous across release sites. This heterogeneity can well explain differential learning-induced plasticity across boutons within single KCs (Bilz et al., 2020; Davidson et al., 2023).

To better resolve the presynaptic localization of Dop1R1 and Dop2R, we turned to “giant” *Drosophila* neurons differentiated from cytokinesis-arrested neuroblasts in culture (Wu et al., 1990). The expanded size of the giant neurons is advantageous for investigating the microanatomy of neurons in isolation (Saito & Wu, 1991; Wu et al., 1990; Yao et al., 2000). Importantly, these giant neurons exhibit characteristics of mature neurons, including firing patterns (Wu et al., 1990; Yao & Wu, 2001; Zhao & Wu, 1997) and acetylcholine release (Yao et al., 2000), both of which are regulated by cAMP and CaMKII signaling (Yao et al., 2000; Yao & Wu, 2001; Zhao & Wu, 1997). In the giant neurons from the *Brp::SNAP* embryos (Kohl et al., 2014), Brp was localized to the terminals of neurites, but rarely in the proximal neurites (**Figure 5B**). Furthermore, we found punctate Brp clusters in the giant neuron terminals (**Figure 5B**), together recapitulating the essential characteristics of the AZ cytomatrix in adult neurons. To confirm if dopamine receptors localize to these presynaptic sites, we generated the giant neurons from embryos carrying the Venus insertion to *Dop1R1* and *Dop2R* (Kondo et al., 2020) together with *Brp::SNAP*. Both Dop1R1 and Dop2R were expressed in the giant neurons and enriched in the same distal axonal segments (**Figure 5C**). A closer investigation revealed that these receptors

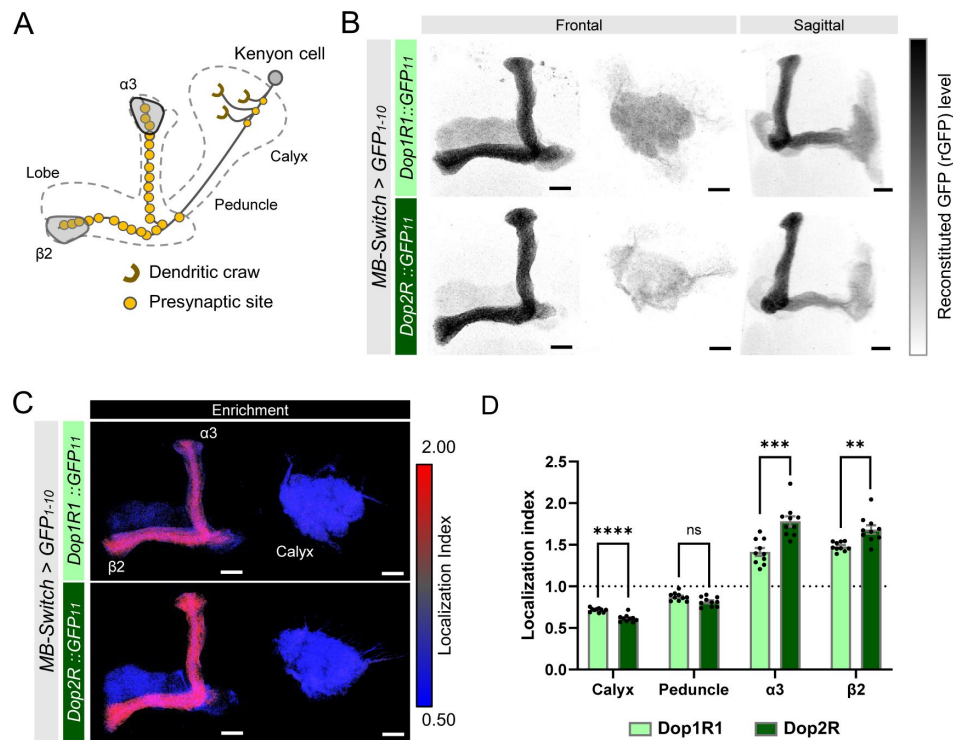


Figure 3.

Subcellular localization of Dop1R1 and Dop2R in the Kenyon cells.

Subcellular localization of Dop1R1 and Dop2R in the KCs is visualized by GFP₁₁-tag. *MB-Switch* was used to express GFP₁₋₁₀ and CD4::tdTomato in the KCs. To activate Gene-Switch, flies were fed with food containing 200 μ M RU486 for 72 hours before dissection.

(A) Schematic showing the projection pattern of a α/β KC.

(B) Enrichment of Dop1R1 and Dop2R in the MB lobe. Maximum-intensity projections of the lobe (left) and the calyx (middle) are shown in frontal view. The whole left MB are shown in sagittal view (right). Reconstituted GFP signals for both Dop1R1::GFP₁₁ and Dop2R::GFP₁₁ distributed throughout the MB lobe and the calyx.

(C) Visualization by LI showed more pronounced enrichment of Dop2R than Dop1R1 in the lobe.

(D) Mean LI of Dop1R1 and Dop2R in the calyx, the peduncle, the $\alpha 3$ and $\beta 2$ compartment in the lobe. Student's t test was performed to compare LI of Dop1R1 and Dop2R in each region (N = 10). Error bars; SEM. p > 0.05, ** p < 0.01, *** p < 0.001, **** p < 0.0001, ns: not significant p > 0.05.

Scalebars, 20 μ m (B and C).

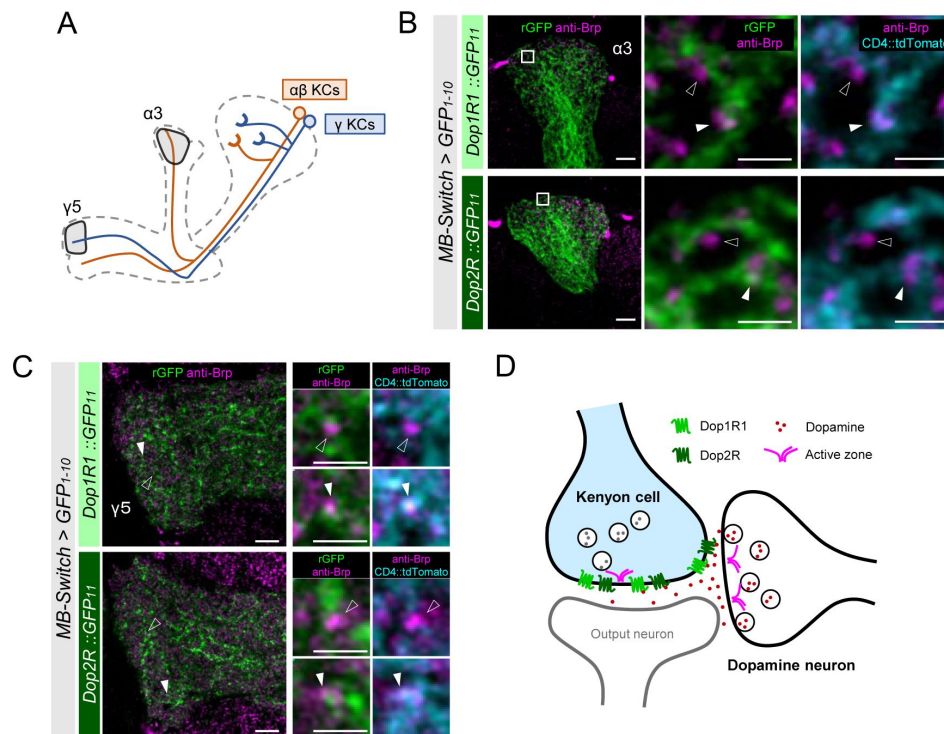


Figure 4.

High-resolution imaging revealed the two opposing dopamine receptors existing on the presynaptic and postsynaptic sites of Kenyon cells.

(A) Schematic showing the projection pattern of α/β and γ KCs.

(B and C) Airyscan images of Dop1R1::rGFP and Dop2R::rGFP in KCs (green) co-labeled with the AZ stained with anti-Brp (magenta). *MB-Switch* was used with 72 hours of RU486 feeding to express GFP₁₋₁₀ and CD4::tdTomato in the KCs. Brp puncta that overlap with CD4::tdTomato signals (cyan) are identified to be presynaptic sites of in KCs (white arrowheads), and those do not overlap are determined to be presynaptic sites of non-KCs (outlined arrows). The synaptic localization of these receptors is similar in the $\alpha 3$ (B) and $\gamma 5$ (C) compartments. In the right panels, white squared regions in the left panels are magnified. Scalebars, 5 μ m (left), 1 μ m (right).

(D) Illustration of localization of Dop1R1 and Dop2R to presynaptic and postsynaptic sites in the axon terminal of KCs.

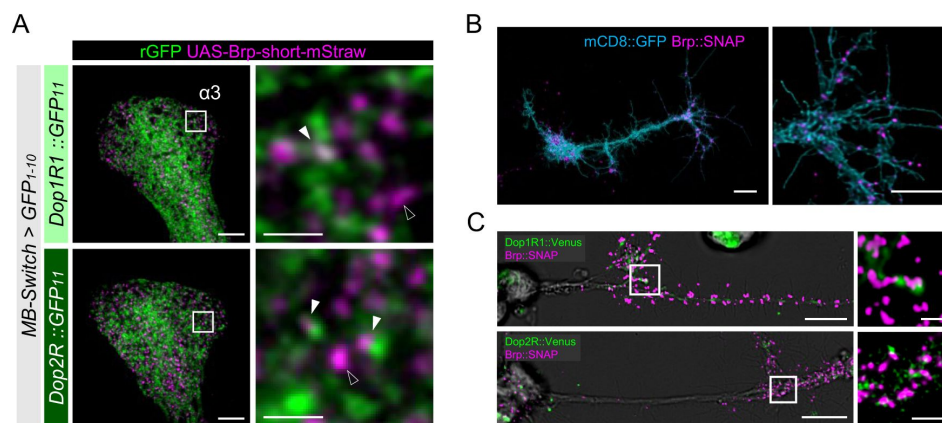


Figure 5.

Presynaptic localization of Dop1R1 and Dop2R in Kenyon cells and giant neurons.

(A) Double labelling of dopamine receptors (green) and the AZ of the KCs (magenta). *MB-Switch* was used with 72 hours of RU486 feeding to express GFP_{1-10} and $Brp^{short}::mStraw$ in the KCs. Single focal slices at the $\alpha 3$ compartment are shown. White squares in the left panels are magnified in the right panels. The Brp puncta in KCs were either abutted by the dopamine receptor signals (white arrowheads) or had barely detectable signals nearby (outlined arrowheads). Scale bars, 5 μm (left), 1 μm (right).

(B) Punctate Brp expression in a giant neuron culture differentiated from cytokinesis-arrested neuroblasts of *OK371-GAL4/UAS-mCD8::GFP* embryos. Aggregated Brp condensates (magenta) were observed in the neurite terminals of the cells marked with *mCD8::GFP* (cyan) in the right panel. Scale bars, 20 μm (left), 10 μm (right).

(C) Double labelling of dopamine receptors (green) and the AZs (magenta). *Dop1R1::Venus* or *Dop2R::Venus* was crossed with *Brp::SNAP*. In the left panels, giant neurons extending their neurites from the cell body on the left to the right. In the right panels, white squared regions in the left panels are magnified. Scale bars, 10 μm (left), 2 μm (right).

are associated with the Brp clusters (**Figure 5C**). These observations in the giant neurons are strikingly similar to those in the KCs (**Figure 3**, **4** and **5A**), corroborating that the presynaptic localization of these receptors is independent of the circuit context.

Distinct subcellular enrichment of dopamine receptors in MBONs and DANs

Presynaptic and postsynaptic localization of Dop1R1 and Dop2R in KCs prompted us to investigate MBONs, that have profound postsynaptic contacts with DANs in the MB compartments (Li et al., 2020; Takemura et al., 2017). MBON- γ 1pedc $\alpha\beta$ (also known as MB-MVP2, MBON-11; Aso, Hattori, et al., 2014; Tanaka et al., 2008) has most of its postsynaptic sites on the γ 1 compartment and the peduncle of the α/β neurons and send axonal projections to the α and β lobes (**Figure 6**; Dorkenwald et al., 2024; Schlegel et al., 2024). We analyzed the subcellular localization of Dop1R1::rGFP and Dop2R::rGFP in MBON- γ 1pedc $\alpha\beta$ by driving GFP₁₋₁₀ using *R83A12-GAL4* (Perisse et al., 2016). Strikingly, both Dop1R1 and Dop2R are enriched in the dendritic projection in the γ 1 compartment (**Figure 6B** and **6C**) in sharp contrast to the sparse distribution of these receptors in the dendrites of KCs (**Figure 3B**–**3D**). This localization is consistent with previous reports of functional DAN>MBON synapses (Takemura et al., 2017) and dopaminergic plasticity on the dendrites (Boto et al., 2019; Pribbenow et al., 2022). Additionally, we detected these receptors in the presynaptic boutons (**Figure 6C**), which are consistent with the results in KCs and in the giant neurons (**Figure 4B**, **4C**, **5A** and **5C**).

D₂-like receptors in mammals are expressed in DANs and act as autoreceptors, which mediates feedback signals by receiving neurotransmitters released from the neuron on which the receptors reside (Ford, 2014). In *Drosophila*, multiple dopamine receptor genes are expressed in DANs (**Figure 1D** and **1E**; Aso et al., 2019; Deng et al., 2019; Kondo et al., 2020), but it is unclear if these receptors function as autoreceptors. We therefore examined the subcellular localization of Dop1R1 and Dop2R in the PAM cluster DANs with a particular focus on their presynaptic terminals. The PAM neurons are polarized; presynaptic proteins are abundantly enriched in the MB lobe and barely detected in dendrites (**Figure 7A** and **7B**). We visualized Dop1R1 and Dop2R proteins in the PAM cluster DANs using *R58E02-GAL4*, and both were localized to the terminals in the MB lobes (**Figure 7C**). Representation of LI for Dop1R1 and Dop2R in the PAM neurons again showed stronger presynaptic enrichment of Dop2R than Dop1R1 (**Figure 7C**). Dop2R was strongly enriched at β '1 compartment showing significantly higher LI than Dop1R1 (**Figure 7D** and **7E**). Higher magnification revealed the accumulation of both Dop1R1 and Dop2R in the boutons (**Figure 7F**). The presynaptic localization of Dop1R1 and Dop2R in DANs suggests that both receptors mediate feedback regulation.

State-dependent and bidirectional modulation of dopamine receptor expression in the PAM and PPL1 DANs

The activity of MB-projecting DANs is reported to be dynamic and sensitive to feeding states (Ichinose et al., 2017; Liu et al., 2012; Plačaiš & Preat, 2013; Senapati et al., 2019; Siju et al., 2020; Tsao et al., 2018; Yamagata et al., 2016). We therefore examined if starvation alters the protein expression of Dop1R1 and Dop2R in these MB-projecting DANs (**Figure 8A**). We found a significant elevation of Dop1R1 in the terminals of PAM- γ 5 upon starvation for 10 hours or longer (**Figure 8B** and **8C**). In contrast, starvation did not increase Dop2R in PAM- γ 5 but rather tended to decrease, if at all (**Figure 8B** and **8C**). We found similar starvation-dependent changes in Dop1R1 and Dop2R levels in other PAM neurons (**Figure 8 – figure supplement 1A–1D**). These results together suggest that starvation enhances presynaptic dopamine signaling in the reward-related PAM neurons by shifting the balance of Dop1R1 and Dop2R.

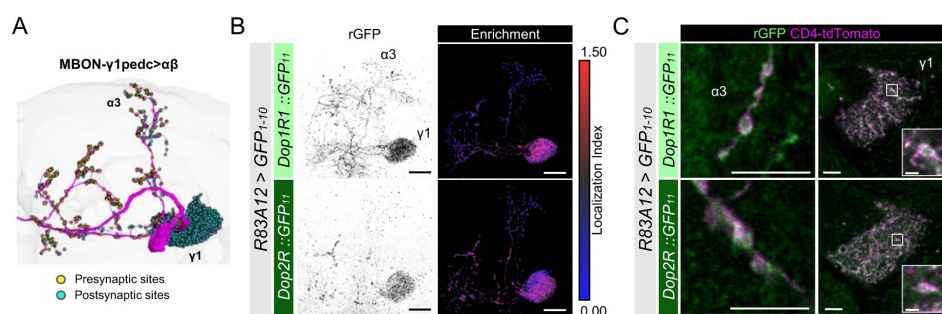


Figure 6.

Subcellular localization of Dop1R1 and Dop2R in MBON-γ1pedc>αβ.

(A) The projection pattern of MBON-γ1pedc>αβ from the tracing data in FlyWire (Dorkenwald et al., 2024; Schlegel et al., 2024).

(B and C) *R83A12-GAL4* was used to express *UAS-GFP₁₋₁₀* and *UAS-CD4::tdTomato* in the PAM neurons.

(B) Reconstituted GFP signals (left) of *Dop1R1::GFP₁₁* and *Dop2R::GFP₁₁* in MBON-γ1pedc>αβ. Maximum-intensity projections of the left MB lobe. Visualization of LI (right) revealing that both Dop1R1 and Dop2R are enriched in the dendritic projection of MBON-γ1pedc>αβ in the γ1 compartment as well as in the presynaptic boutons.

(C) Airyscan images of the presynaptic boutons around α3 (left) and dendritic projections in the γ1 compartment (right). White squares in the right panels are magnified in the insertion to show the swelling membrane structures with punctate localization of dopamine receptors.

Scale bars, 20 μm (B), 5 μm (C), 1 μm (C, insertion).

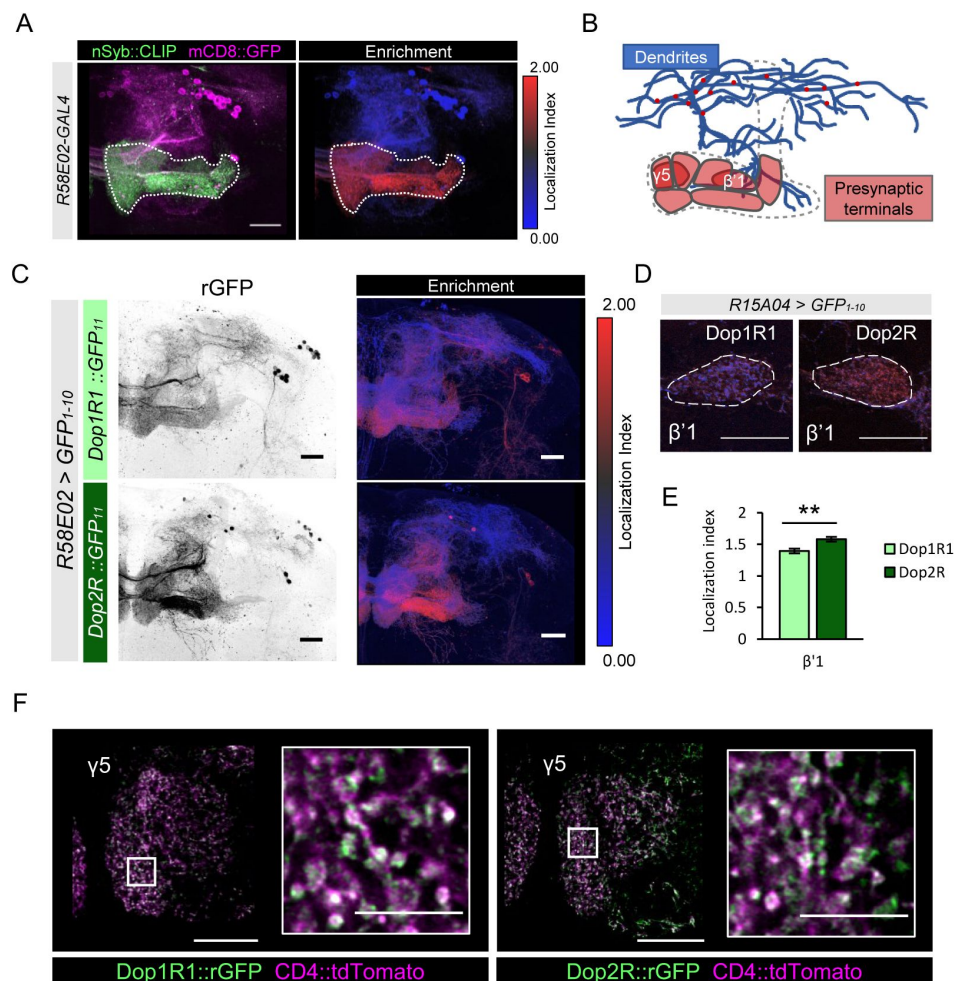


Figure 7.

Subcellular localization of Dop1R1 and Dop2R in dopamine neurons.

(A) Maximum-intensity projection image showing the distribution of presynaptic sites in the PAM neurons. Left panel: *R58E02-GAL4* was used to express *mCD8::GFP* (magenta) and *nSyb::CLIP* (magenta). Right panel: Visualization by LI showing enrichment of *nSyb* signals in the lobe projection of the PAM neurons.

(B) Illustrated projection pattern of the PAM neurons. Red puncta on the dendrites indicate the sparse distribution of presynaptic sites in dendrites.

(C-F) Subcellular localization of GFP₁₁-tagged Dop1R1 and Dop2R in the PAM neurons. *R58E02-GAL4* (C and F) or *R15A04-GAL4* (D and E) was used to express *UAS-GFP₁₁₋₁₀* and *UAS-CD4::tdTomato* in the PAM neurons.

(C) Reconstituted GFP signals of *Dop1R1::GFP₁₁* and *Dop2R::GFP₁₁* in PAM neurons (left). LI visualization revealed the stronger presynaptic enrichment of Dop2R than that of Dop1R1 (right). Maximum-intensity projections of the left hemisphere including the whole MB lobe and dendritic projections of the PAM neurons around the MB.

(D and E) LI in PAM-β'1 neuron. (D) The presynaptic terminals of PAM-β'1 neurons are shown (dashed line). (E) Mean LI for Dop1R1 and Dop2R in the β'1 (Mann-Whitney U test, N = 9). Error bars; SEM.

(F) A single optical slice of the γ5 compartment in the MB lobe obtained using Airyscan. Merged image of reconstituted GFP (green) and *CD4::tdTomato* (magenta). Insertions are the magnified images of the presynaptic boutons of PAM-γ5 (white squares).

Scale bars, 20 μm (A, C, D and F), 5 μm (F, insertion).

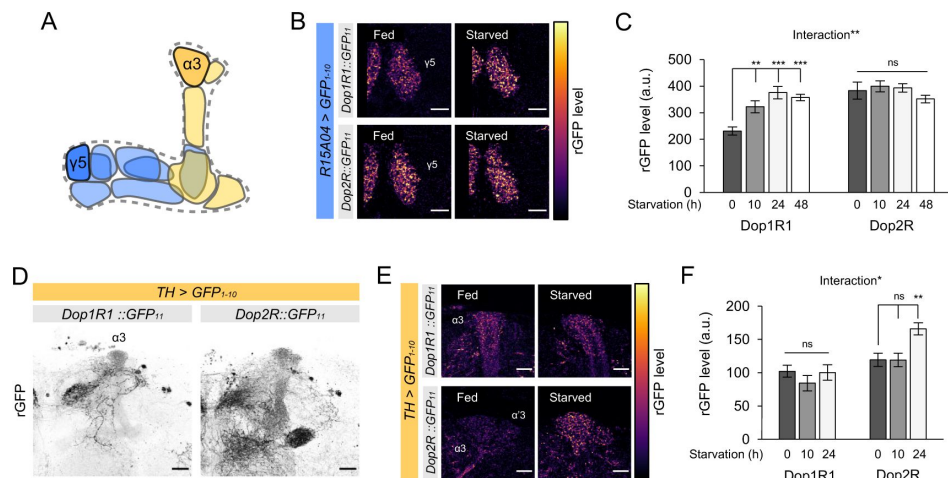


Figure 8.

Bidirectional modification of dopamine receptor expression in dopamine neurons.

(A) Schematic illustration of the MB projection of the PAM and PPL1 dopamine neurons.

(B) Dop1R and Dop2R in the presynaptic terminals of PAM-y5 after 48 hours of starvation compared with fed state.

(C) Quantification of rGFP signal levels in the presynaptic terminals of PAM-y5 after 0, 10, 24 and 48 hours of starvation (n = 6-13).

(D) Reconstituted GFP signals of *Dop1R1::GFP₁₋₁₀* and *Dop2R::GFP₁₁* in the PPL1 neurons. In the MB projections of the PPL1 neurons, Dop1R1 was detected in only the α3 compartment. Dop2R was found in all MB projections. Maximum-intensity projections of the MB lobe.

(E) Dop1R and Dop2R in the presynaptic terminals of PPL1-α3 after 24 hours of starvation compared with fed state.

(F) Quantification of rGFP signal levels in the presynaptic terminals of PPL1-α3 after 0, 10 and 24 hours of starvation (n = 7-10). Scale bar, 10 μm (B and E), 20 μm (D).

Interaction effects between genotypes and starvation time on protein levels were tested by Two-way ANOVA (C and F). Bars and error bars represent mean and SEM, respectively (C and F). ** p < 0.01, *** p < 0.001, ns: not significant p > 0.05.

DANs of the PAM and PPL1 clusters exert distinct, largely opposite behavioral functions (Claridge-Chang et al., 2009; Liu et al., 2012). Therefore, plasticity in the PPL1 neurons may differ from that in the PAM neurons. To test this hypothesis, we examined the starvation-dependent change in Dop1R1 and Dop2R protein expression in the PPL1 neurons. To this end, we visualized GFP₁₁-tagged receptors by expressing GFP₁₋₁₀ using *TH-GAL4* (Friggi-Grelin et al., 2003). We detected Dop2R proteins in all MB projections of the PPL1 neurons, whereas Dop1R1 proteins were only detectable in the terminals of the PPL1- α 3 neuron (Figure 8D). Strikingly, the starvation-induced changes in the PPL1 neurons were opposite to those in the PAM: Dop2R, but not Dop1R1, was significantly increased in the α 3 compartment (Figure 8E and 8F). In the other compartments, the Dop2R::rGFP levels tended to be higher in starved flies although the increase was not statistically significant (Figure 8 – figure supplement 1E and 1F). These results are in line with the state-dependent changes in the physiology of these DANs (Siju et al., 2020; Tsao et al., 2018). Taken together, starvation induces bidirectional modulation of the dual autoreceptor system in the PPL1 and PAM DANs, and we propose that these changes shift the balance of dopamine output from these subsets (Figure 9A).

The dopaminergic signals from the PPL1 neurons inhibit expression of appetitive olfactory memory in fed flies (Krashes et al., 2009; Pavlovsky et al., 2018). The increased Dop2R autoreceptors in starved flies (Figure 8E and 8F) may thus disinhibit appetitive memory expression by suppressing PPL1 outputs (Figure 9B). To test this hypothesis, we examined appetitive memory by transgenic knockdown of Dop2R specifically in the PPL1 cluster neurons using *MB504B-GAL4* (Vogt et al., 2014). Indeed, appetitive memory was impaired (Figure 9C), suggesting the significance of the negative feedback regulation of the PPL1 via Dop2R in appetitive behavior expression. This result is consistent with our model, which suggests that enhanced negative feedback to PPL1 via Dop2R in the starved state contributes to the expression of appetitive behavior (Figure 9A and 9B).

Discussion

The present study demonstrated the expression, subcellular localization, and dynamics of endogenous dopamine receptors, specifically Dop1R1 and Dop2R, in the MB circuit of the fly brain. Compared to intronic insertions of reporters using transposons (Fendl et al., 2020), CRISPR/Cas9-based insertion is advantageous in terms of flexibility in fusion sites (Bonanno et al., 2024; R. Kamiyama et al., 2021; Sanfilippo et al., 2024; Williams et al., 2019). This enables the C-terminal tagging of dopamine receptors that do not perturb protein localization and function (Figure 2 – figure supplement 1).

The split-GFP reconstitution strategy enables sensitive detection of low-abundant endogenous proteins by the tandem multimerization of the GFP₁₁ tags, as shown in other epitope tags such as SunTag and smFPs (Tanenbaum et al., 2014; Viswanathan et al., 2015). The native fluorescence of rGFP was sufficient to detect tagged proteins at the high signal-to-noise ratio, without the need for signal amplification using antibodies. The background fluorescence of the split-GFP fragments was practically negligible (Figure 8D; D. Kamiyama et al., 2016; Kondo et al., 2020). This approach can thus be applicable to monitor localization and dynamics of endogenous proteins of low abundance.

Our cell-type-specific receptor labeling offers further potential for ultrastructural examination using higher resolution imaging techniques. For example, synaptic condensates of Dop1R1 and Dop2R at the resolution in this study (Figures 4 and 5) did not allow us to distinguish the receptors on the plasma membrane and those that are internalized (Dumartin et al., 1998; Kotowski et al., 2011). Thus, application of super-resolution fluorescent imaging and expansion microscopy (Gao et al., 2019) would disentangle the precise membrane localization of the receptors.

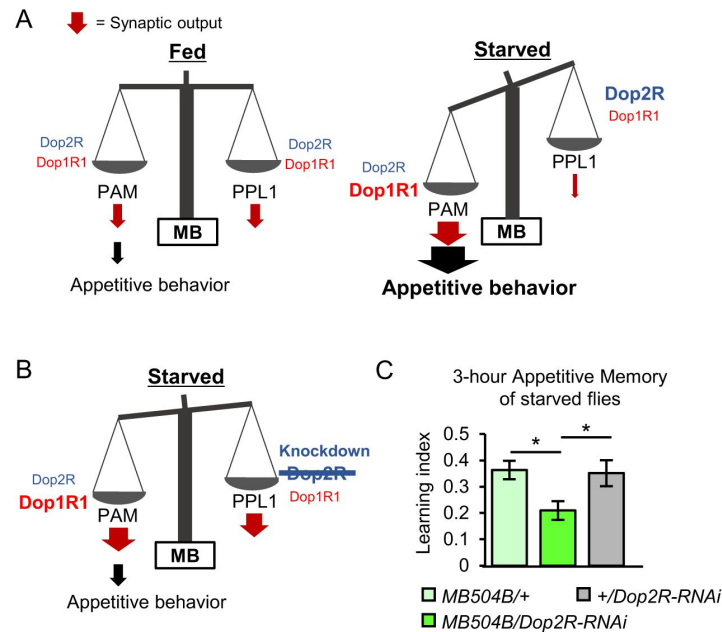


Figure 9.

The dual dopaminergic feedback regulating starved-state dependent expression of appetitive behavior.

(A) A working model showing the role of the dual dopaminergic feedback regulation. In the starved state, increased Dop1R1 in PAM neurons and increased Dop2R in PPL1 neurons changes the balance between the synaptic outputs from these DANs to favor appetitive behavior.

(B) According to the model, loss of Dop2R in PPL1 upregulates output from PPL1 to attenuate appetitive behavior in starved flies.

(C) Knockdown of Dop2R in the PPL1 neurons by *MB504B-GAL4* reduced 3-hour appetitive memory performance (t-test with Bonferroni correction, $n=14-15$). Bars and error bars represent mean and SEM, respectively. * $p < 0.05$.

Spatial regulation of dopamine signaling through the two opposing receptors

How do Dop1R1 and Dop2R function in presynaptic terminals? Released dopamine is not confined to the synaptic cleft, but diffused in the extracellular space (Liu et al., 2021 [↗](#); Rice & Cragg, 2008 [↗](#)). Such volume transmission can increase the number of target synapses by 10 times (Li et al., 2020 [↗](#); Takemura et al., 2017 [↗](#)). Thus, the broader distribution of Dop1R1 (**Figure 3B** [↗](#)-**3D** [↗](#) and **7C** [↗](#)-**7E** [↗](#)) may enable it to respond to extrasynaptic dopamine. In response to residual dopamine, Dop2R at the AZ may inhibit activities of adenylate cyclases and voltage-gated calcium channels (Beaulieu & Gainetdinov, 2011 [↗](#); Hearn et al., 2002 [↗](#)), thereby reducing the noise of the second messengers. Collectively, the spatial configuration of Dop1R1 and Dop2R may enhance the sensitivity and precision of dopaminergic modulation.

Furthermore, presynaptic receptor localization (**Figure 4B** [↗](#)-**4D** [↗](#), **5A** [↗](#), **5C** [↗](#), **6C** [↗](#) and **7F** [↗](#)) suggests a spatially confined cAMP, forming nanodomains (Anton et al., 2022 [↗](#); Bock et al., 2020 [↗](#); Maiellaro et al., 2016 [↗](#); Zhang et al., 2020 [↗](#)). cAMP signaling has been shown to regulate multiple events at presynaptic terminals, such as the molecular assembly at the AZ and synaptic vesicle dynamics (Ehmann et al., 2018 [↗](#); Kittel & Heckmann, 2016 [↗](#); Kuromi & Kidokoro, 2000 [↗](#), 2005 [↗](#); Maiellaro et al., 2016 [↗](#); Renger et al., 2000 [↗](#); Sachidanandan et al., 2023 [↗](#)). Especially in KCs, cAMP-dependent plasticity underlies associative memory (Boto et al., 2014 [↗](#); Cohn et al., 2015 [↗](#); Louis et al., 2018 [↗](#); Noyes & Davis, 2023 [↗](#); Stahl et al., 2022 [↗](#)). Presynaptic cAMP regulation through Dop1R1 and Dop2R therefore explains the requirement of these receptors in associative memory (Kim et al., 2007 [↗](#); Scholz-Kornehl & Schwärzel, 2016 [↗](#)).

The dual autoreceptor system may shape dopamine release

Presynaptic localization of Dop1R1 and Dop2R in DANs (**Figure 7C** [↗](#)-**7F** [↗](#)) strongly suggests their functions as autoreceptors. In the *Drosophila* nervous system, Dop2R was shown to negatively regulate dopamine release (Shin & Venton, 2022 [↗](#); Vickrey & Venton, 2011 [↗](#)). Tight presynaptic enrichment in PAM (**Figure 7F** [↗](#)) suggests that Dop2R receives high levels of dopamine and effectively prevents overactivation. The presence of Dop1R1 in DAN terminals (**Figure 7F** [↗](#)) was unexpected and introduces another layer of presynaptic regulations to dopamine release. Although it must be functionally verified, presynaptic Dop1R1 likely provides positive feedback to the dopamine release as an autoreceptor. This positive feedback would be particularly important for signal amplification when extracellular dopamine concentrations are low. Consistent this hypothesis, presynaptic Dop1R1 was undetectable in most PPL1 DANs (**Figure 8D** [↗](#)), which were reported to have high spontaneous activities (Feng et al., 2021; Plačaiš & Preat, 2013 [↗](#)). Taken together, this dual autoreceptor system likely fine-tune the amplitude and kinetics of dopamine release. Alternatively, these presynaptic receptors could potentially receive extrasynaptic dopamine released from other DANs. Therefore, the autoreceptor functions need to be experimentally clarified by manipulating the receptor expression in DANs.

Dopamine receptor expression is reported to be associated with prolonged exposure to psychoactive substances, such as caffeine and ethanol (Andretic et al., 2008 [↗](#); Kanno et al., 2021 [↗](#); Kondo et al., 2020 [↗](#); Petrucci et al., 2018 [↗](#)). Our study further showed starvation-dependent changes of Dop1R1 and Dop2R in DAN terminals (**Figure 8** [↗](#)). Strikingly, starvation responses of presynaptic Dop1R1 and Dop2R were differential depending on the DAN cell types (**Figure 8** [↗](#)). These results indicate that starvation bidirectionally changes the dual autoreceptor system, putting more weight on the PAM output over PPL1 to control the expression of appetitive behavior (**Figure 9** [↗](#)). This shifted balance explains the state-dependent changes in the presynaptic activity of these two clusters of DANs (Siju et al., 2020 [↗](#); Tsao et al., 2018 [↗](#)) and bidirectional modulation of the MB output (Aso, Sitaraman, et al., 2014 [↗](#); Ichinose et al., 2021 [↗](#); Oswald et al., 2015 [↗](#)).

Materials and Methods

Key resource table

	Source	Identifier
Antibodies and reagents		
Mouse anti-TH	ImmunoStar Inc., Hudson, WI, USA	#22941
Mouse anti-Brp	Developmental Studies Hybridoma Bank, Iowa city, IA, USA	nc82
Rabbit anti-DsRed	Takara Bio USA, Inc., San Jose, CA, USA	#632496
Goat anti-mouse Alexa Fluor 405	Invitrogen, Waltham, MA, USA	#A31553
Goat anti-mouse Alexa Fluor 633	Invitrogen, Waltham, MA, USA	#A21052
Goat anti-rabbit Alexa Fluor 568	Invitrogen, Waltham, MA, USA	#A11036
SNAP-Cell 647-SiR	Ipswich, MA, USA	S9102S
CLIP-Surface 547 substrate	New England Biolabs Inc., Ipswich, MA, USA	#S9233S
RU486 (Mifepristone)	Tokyo Chemical Industry Co., Tokyo, Japan	#M1732
<i>Drosophila</i> strain		
<i>Dop1R1-T2A-GAL4</i>	Kondo et al., 2020	N/A
<i>Dop2R-T2A-GAL4</i>	Kondo et al., 2020	N/A
<i>Dop1R1-T2A-LexA</i>	Kondo et al., 2020	N/A
<i>Dop1R1::7xGFP₁₁</i>	Kondo et al., 2020	N/A
<i>Dop2R::7xGFP₁₁</i>	Kondo et al., 2020	N/A
<i>DopEcR::7xGFP₁₁</i>	Kondo et al., 2020	N/A
<i>Dop1R1::Venus</i>	Kondo et al., 2020	N/A
<i>Dop2R::Venus</i>	Kondo et al., 2020	N/A
<i>UAS-GFP₁₋₁₀</i>	Kondo et al., 2020	N/A
<i>UAS-mCD8::GFP</i>	Pfeiffer et al., 2010	BDSC #32194
<i>lexAop-rCD2::GFP</i>	Miyamoto et al., 2012	N/A
<i>UAS-CD4::tdTomato</i>	C. Han et al., 2011	BDSC #35841
<i>UAS-nSyb::CLIP</i>	Kohl et al., 2014	BDSC #58398
<i>UAS-Brp^{short}::mStraw</i>	Fouquet et al., 2009	N/A
<i>Brp::SNAP</i>	Kohl et al., 2014	BDSC #58397
<i>MB-Switch</i>	Mao et al., 2004	BDSC #81013
<i>OK371-GAL4</i>	Mahr & Aberle, 2006	BDSC #26160
<i>R83A12-GAL4</i>	Jenett et al., 2012	BDSC #40348

<i>R15A04-GAL4</i>	Janett et al., 2012	BDSC #48671
<i>R58E02-GAL4</i>	Jenett et al., 2012	BDSC #41347
<i>TH-GAL4</i>	Friggi-Grelín et al., 2003	N/A
<i>MB504B</i>	Aso, Hattori et al., 2014	BDSC #68329
<i>UAS-Dop2R-RNAi</i>	Perkins et al., 2015	BDSC #50621
Software		
Fiji	Schindelin et al., 2012, http://fiji.sc	RRID: SCR_002285
GraphPad Prism 5	GraphPad Software	RRID: SCR_002798

Flies

Flies were raised on standard cornmeal food at 25°C under a 12:12 h light-dark cycle (ZT0 at 8 AM) for all experiments. The GAL4-UAS system was used to express the transgenes of interest in specific neuron subtypes. Flies carrying GAL4 were crossed to another strain carrying UAS reporters, and F1 progenies were used in the experiments. To visualize GFP₁₁-tagged dopamine receptors in the specific cell-types, female fly strains carrying *UAS-CD4::tdTomato*, *UAS-GFP₁₋₁₀* and GAL4 driver were crossed to male fly strains carrying *Dop1R1::GFP₁₁* or *Dop2R::GFP₁₁*, and F1 progenies were used. To make the giant neuron culture, embryos from *Brp::SNAP* or the F1 progeny of *OK371-GAL4* crossed with *Brp::SNAP*, *UAS-mCD8::GFP*, *UAS-nSyb::CLIP* flies were used. For the fly lines used in our manuscript, see Key resource table.

RU486 feeding

To activate Gene-Switch, flies are fed with food containing RU486 (mifepristone). Food containing 200 µM of RU486 was prepared as described previously (Mao et al., 2004 [\[4\]](#)). In brief, 200 mg of RU486 was dissolved in 10 ml of 99.5% ethanol to make a stock solution. 4.3 µl of the stock solution was added to 1 ml of molten fly food and mixed thoroughly. The molten food with RU486 was poured into vials or bottles and cooled to make solid food.

Brain dissection and Immunohistochemistry

Flies were sorted under CO₂ anesthesia to select males with the specific genotypes and kept in a food vial for recovery at least 24 h prior to the dissection. Fly brains were dissected 3-7 days after eclosion in ice-cold Phosphate-buffered saline (PBS). After dissection, brains were kept in ice-cold PBS with 2% paraformaldehyde (PFA) for up to 30 min. For fixation, brains were incubated in 2 % PFA in PBS for 1 hour at room temperature. Fixed brains were washed in PBS with 0.3 % Triton X-100 (PBT) for 10 min 3 times. For rGFP imaging, fixed brains were mounted in SeedB2 (Ke et al., 2016 [\[4\]](#)), and native fluorescence was imaged without signal amplification.

For chemical tagging reaction of nSyb::CLIP, brains were incubated in PBT containing CLIP-Surface 547 substrate (1µM) for 15 min at room temperature. Brains were washed in PBT for 10 min 3 times.

For immunohistochemistry, fixed brains were blocked in 3 % normal goat serum (NGS; Sigma-Aldrich, G9023) for 30 min at room temperature. Brains were incubated in primary and secondary antibodies diluted in PBT with 1 % NGS over two nights at 4 °C, respectively. After each step, brains were washed three times in PBT for longer than 20 min at room temperature and mounted in 86% glycerol. The following primary antibodies were used: mouse anti-TH (1:100), mouse anti-Brp (1:40), rabbit anti-DsRed (1:2000). Secondary antibodies: Alexa Fluor 633 goat anti-mouse (1:400 for anti-TH, 1:200 for anti-Brp).

Fluorescent imaging

For image acquisition, whole-mount brains were scanned with the Olympus FV1200 confocal microscope with following objective lens; 20x oil (NA = 0.85, UPLSAPO20XO, Olympus; [Figure 1A](#), [1B](#), [1G](#) and [1H](#)), 30x silicone (NA = 1.05, UPLSAPO30XS, Olympus; [Figure 3B](#) and [3C](#)), 40x oil (NA = 1.3, UPLFLN40XO, Olympus; [Figure 2B](#), [2C](#), [6B](#), [7A](#), [7C](#), [7D](#), [8B](#), [8E](#), [Figure 8 – figure supplement 1-A,C,E](#)), or 60x oil (NA = 1.42, PLAPON60XO, Olympus; [Figure 1D](#), [1E](#), [1I](#), [Figure 1 – figure supplement 1](#)). Z-stack images were acquired.

For high resolution imaging in [Figure 4B](#), [4C](#), [5A](#), [6C](#), [7F](#) and [Figure 5 – figure supplement 1](#), we used Airyscan on Zeiss LSM800 with 63x oil objective lens (NA = 1.40, Plan-Apochromat).

Image analysis

All image analysis were conducted on Fiji ([Schindelin et al., 2012](#)).

To visualize the subcellular localization of dopamine receptors in defined neurons, we devised the localization index (LI). Relative receptor density was calculated in each voxel by dividing the receptor (reconstituted GFP) signals by the corresponding membrane signals (CD4::tdTomato). To set ROI, any voxels devoid of membrane signal were censored. The local density was normalized by the mean values in the ROI. For the visual representation, the reference membrane signals were colored according to the normalized LI ([Figure 2C](#)). To quantify mean LI in each subcellular region ([Figure 3H](#), [7E](#)), ROI was manually set.

Embryonic giant neuron culture

Multinucleated giant neurons from neuroblasts were generated as described previously ([Wu et al., 1990](#)). Briefly, the interior of a gastrula from stage 6/7 embryo was extracted with a glass micropipette and dispersed into a drop of culture medium (~ 40 μ m) on an uncoated coverslip. The culture medium contained 80% *Drosophila* Schneider's insect medium (S0146-100ML, Merck KGaA, Darmstadt, Germany) and 20% fetal bovine serum (F2442-100ML, Merck KGaA, Darmstadt, Germany), with the addition of 50 μ g/ml streptomycin, 50 U/ml penicillin (all from Sigma, St. Louis, MO, USA) and cytochalasin B (CCB; 2 μ g/ml; Sigma, St. Louis, MO, USA). CCB was removed by washing with CCB-free medium 1 d after plating. All cultures were grown in a humidified chamber.

To label Brp::SNAP, cells were incubated in fluorescent SNAP substrate diluted in the culture medium (1:5000, SNAP-Cell 647-SiR) for half an hour at RT. Subsequently, the same cells were labeled by the SNAP substrate to minimize the crosstalk. After the incubation, cells were washed with the culture medium and subjected to confocal scanning with Leica TCS SP8 microscopy equipped with a 40x oil immersion objective (HC PL APO 40x/1.30 Oil PH3 CS2, Leica). Acquired images were then processed with the LIGHTNING software.

Behavioral assays

The experimental protocols for appetitive learning experiment ([Figure 9C](#)) were as described previously ([Ichinose and Tanimoto., 2016](#)). For appetitive conditioning, a group of approximately 50 flies in a training tube alternately received 3-octanol (3OCT; Merck) and 4-methylcyclohexanol (4MCH; Sigma-Aldrich), for 1 minute in a constant air flow with or without reward with an interval of 1 minute between the two odor presentations. These odorants were diluted to 1.2% and 2% with paraffin wax (Sigma-Aldrich), respectively. Dried 2 M sucrose (Sigma-Aldrich) on a piece of filter paper was used as the reward. Flies were starved in the presence of wet tissue paper for 24 hours before appetitive conditioning. For testing, flies were given a choice between the odor paired with reward (Conditioned stimulus, CS+) and another one unpaired (CS-).

Their distribution in the plexiglass T-maze was video-recorded one frame per second for 2 minutes with the CMOS cameras (GS3-U3-51S5M, FLIR) under infrared LED illumination (Ichinose & Tanimoto, 2016 [DOI](#)). Flies were forced to stay on the floor by applying Fluon (Insect-a-Slip PTFE30, BioQuip Products) on the side and top of T-maze arms.

Learning index was then calculated by counting the number of flies in each arm using ImageJ macro as described previously (Ichinose and Tanimoto, 2016 [DOI](#)) with the following formula (Tempel et al., 1983 [DOI](#)):

$$\text{Learning index} = \frac{\#_{CS-} - \#_{CS+}}{\#_{CS+} + \#_{CS-}}$$

where $\#_{CS+}$ and $\#_{CS-}$ imply the number of flies in the CS+ and the CS-arms, respectively. Learning index was calculated for every second and was averaged for the last 60 s of the 120 s test session. An average of a pair of reciprocally trained groups was used as a single data point.

In aversive associative learning experiment (**Figure 2 – figure supplement 1** [DOI](#)), flies were conditioned with twelve 1.5 second pulses of 90V electric shock paired with either 4MCH or 3OCT (CS+) for 1 minute. After 1 minute of interval, another odor (CS-) was presented without electric shock for 1 minute. Flies were tested immediately after training. In the test, flies were allowed to choose between the CS+ and CS-odors in a T-maze. The number of flies in each side of the arm ($\#_{CS+}$ and $\#_{CS-}$) after 2 minutes was used to calculate learning index as described above.

Statistics

For multiple comparison (**Figure 2 – figure supplement 1** [DOI](#), **Figure 8C** [DOI](#), **8F** [DOI](#) and **Figure 8 – figure supplement 1** [DOI](#) and **Figure 9C** [DOI](#)), statistics were performed by Prism5 (GraphPad). Data were always first tested for normality (Shapiro-Wilk test) and for homoscedasticity (Spearman's test or Bartlett's test). If these assumptions were not violated, parametric tests (one- or two-way ANOVA, followed by Dunnett's test or t-test with Bonferroni correction) were applied. If data did not suffice the assumptions for parametric tests, nonparametric tests (Kruskal-Wallis, followed by Dunn's post hoc pairwise test) were performed. For comparison of two groups, Student's t test (**Figure 3D** [DOI](#) and **Figure 3 – figure supplement 1** [DOI](#)) or Mann-Whitney U test (**Figure 7E** [DOI](#)) were performed.

Bars and error bars represent means and SEM, respectively, in all figures. For all figures, significance corresponds to the following symbols: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; ns: not significant $p > 0.05$.

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Author contributions

H.T. and S.H. conceived the project.

H.T. S.H. and N.Y. designed experiments and interpreted results.

S.K. and H.K. designed, generated, and provided transgenic flies.

H.T and C-F. W. supervised experiments.

S.H., K.S. and N.Y. conducted experiments and analyzed data.

H.T., and S.H. drafted the manuscript.

All authors wrote and revised the manuscript.

All authors read and approved the final manuscript.

Figure supplements

Figure 1 – figure supplement 1.

Co-expression of Dop1R1 and Dop2R genes in Kenyon cells.

Double labelling of *Dop1R1-T2A-LexA* and *Dop2R-T2A-GAL4* expressions by *lexAop-rCD2::GFP* (green) and *UAS-CD4::tdTomato* (red), respectively. Cell bodies of Kenyon cells are shown. A single optical section is shown. Scale bars, 5 μ m.

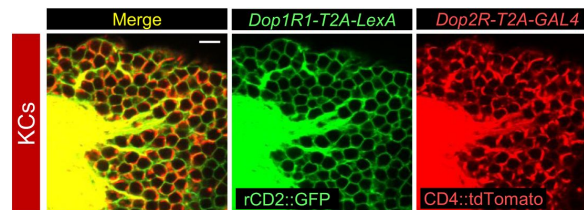


Figure 2 – figure supplement 1.

GFP₁₁-tagging on Dop1R1 and Dop2R do not affect aversive olfactory memory.

To test the functionality of the knock-in alleles, memory of flies carrying homozygous 7xGFP₁₁-tagged Dop1R1 and Dop2R were compared with those without GFP₁₁ tags. To induce reconstitution of GFP in Kenyon cells, GFP₁₋₁₀ was expressed under the control of *MB-Switch*. To activate Gene-Switch, flies were fed with RU486 for 3 days before training. Box plots represent the median as the center line, the upper and lower quartiles as the box boundaries, the minimum and maximum values as the whiskers. Dunn's test was performed (N = 9-12). ns: not significant $p > 0.05$.

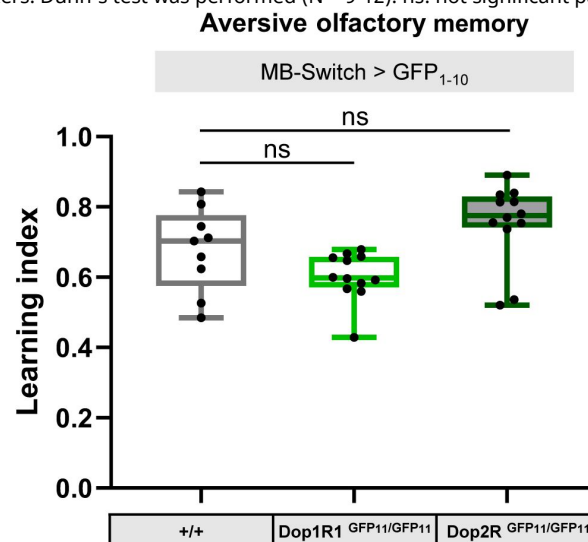


Figure 3 – figure supplement 1.

Localization index remains stable across different experimental batches.

Two experimental batches showed consistent LI for both *Dop1R1::GFP₁₁* and *Dop2R::GFP₁₁* measured in any region in the MB. These two batches were conducted on different days and at different times (Zeitgeber time 5 and 9). The same data set was used in [Figure 3D](#).

Student's t test was performed to compare the two batches (N = 5). Bars and error bars represent mean and SEM, respectively. ns: not significant $p > 0.05$.

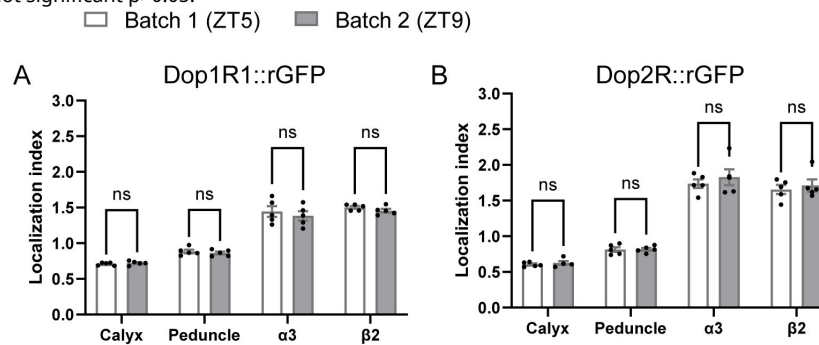
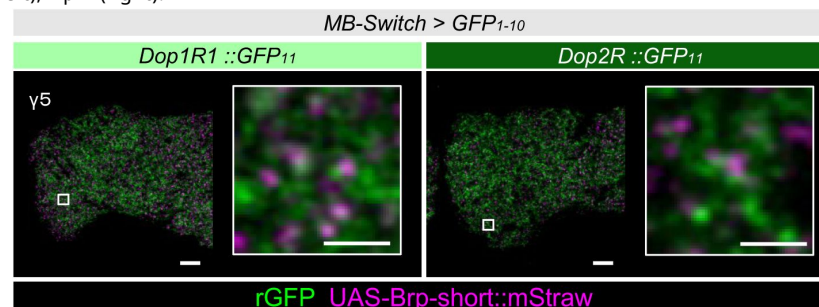


Figure 5 – figure supplement 1.

Presynaptic localization of Dop1R1 and Dop2R in γ Kenyon cells.

Double labelling of dopamine receptors (green) and the AZ of the KCs (magenta). As in the $\alpha 3$ compartment ([Figure 5A](#)), both Dop1R1 and Dop2R were found near the AZ of KCs in the $\gamma 5$ compartment.

Scale bars, 5 μ m (left), 1 μ m (right).



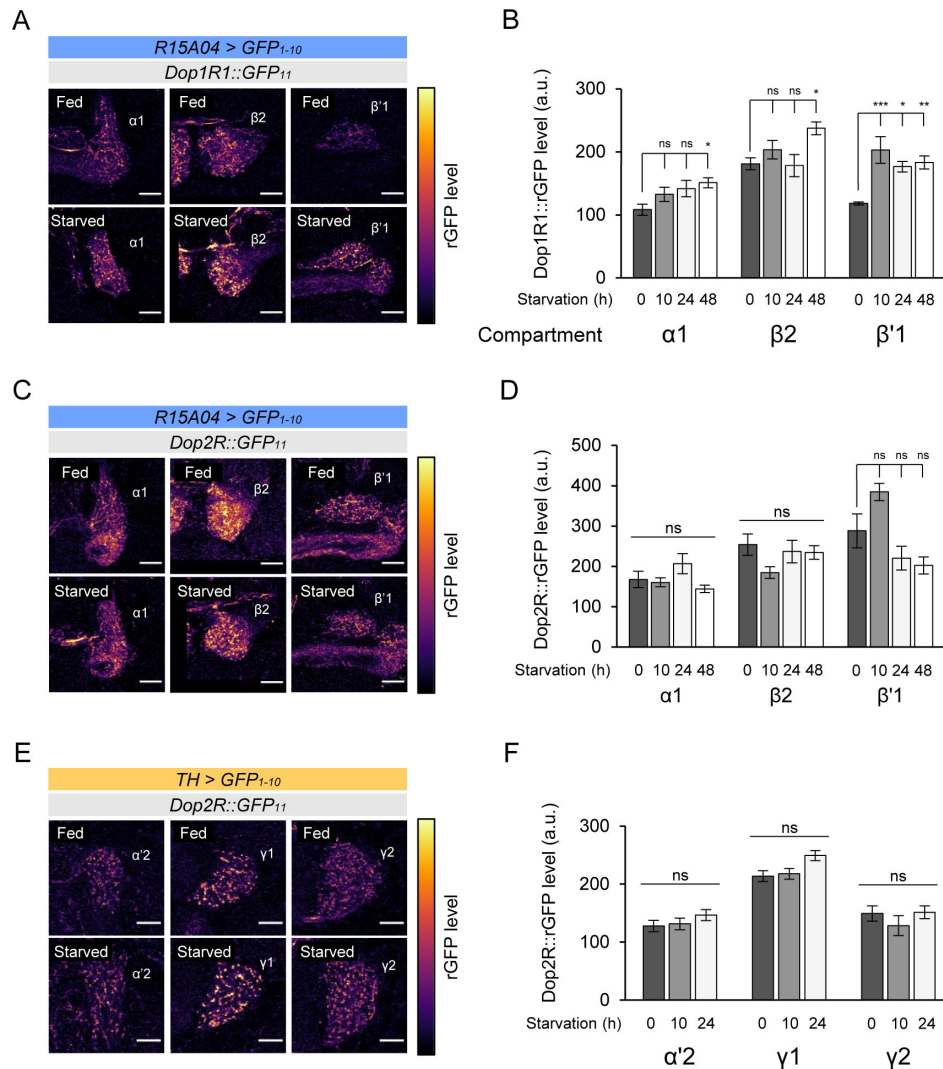


Figure 8 – figure supplement 1.

Starvation-dependent change of dopamine receptors in PAM and PPL1.

(A and C) Dop1R (A) and Dop2R (C) in the presynaptic terminals of PAM- $\alpha 1$, PAM- $\beta 2$ and PAM- $\beta'1$ after 48 hours of starvation compared with fed state.

(B and D) Quantification of dopamine receptor levels in the presynaptic terminals of the PAM neurons after 0 (fed), 10, 24 and 48 hours of starvation (n = 6-13).

(E) Dop2R in the presynaptic terminals of PPL1- $\alpha'2$, PPL1- $\gamma 1$ pedc and PPL1- $\gamma 2$ and after 24 hours of starvation compared with fed state.

(F) Quantification of Dop2R levels in the presynaptic terminals of the PPL1 neurons after 0 (fed), 10 and 24 hours of starvation (n = 7-10).

Scale bar, 10 μ m (A, C, E). Bars and error bars represent mean and SEM, respectively (B, D, F). * p < 0.05, ** p < 0.01, *** p < 0.001, ns: not significant p > 0.05.

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Reviewer #1 (Public review):

Summary:

This is an important and interesting study that uses the split-GFP approach. Localization of receptors and correlating them to function is important in understanding the circuit basis of behavior.

Strengths:

The split-GFP approach allows visualization of subcellular enrichment of dopamine receptors in the plasma membrane of GAL4-expressing neurons allowing for high level of specificity.

The authors resolve the presynaptic localization of DopR1 and Dop2R, in "giant" *Drosophila* neurons differentiated from cytokinesis-arrested neuroblasts in culture as its not clear in the lobes and calyx.

Starvation induced opposite responses of dopamine receptor expression in the PPL1 and PAM DANs provides key insights into models of appetitive learning.

Starvation induced increase in D2R allows for increased negative feedback that the authors test in D2R knockout flies where appetitive memory is diminished.

This dual autoreceptor system is an attractive model for how amplitude and kinetics of dopamine release can be fine tunes and controlled depending on the cellular function and this paper presents a good methodology to do it and a good system where dynamics of dopamine release can be tested at the level of behavior.

Weaknesses:

Key weaknesses have been resolved:

1. Receptor expression is consistent between time of the day and the authors picked two time points. The authors mention that the states of animals could affect LI (e.g. feeding state and anesthesia for sorting, see methods) were kept constant. These data and discussion are helpful.
2. Giant fiber system is argued to be a great model and authors have added additional references. However I am not very deeply familiar with these references or the giant fiber system so I am not completely clear but the argument seems reasonable.
3. The revised manuscript, shows data in the γ KCs (Figure 4C, Figure 5 - figure supplement 1) in addition to α/β KCs, so it appears there is consistency between lobes.
4. The new data for Dop1R1 and Dop2R in MBON- γ 1pedc> $\alpha\beta$ helps with thinking about dopamine receptor co-localization and it would be a herculean task to do this for all the regions but still keeps room open for different scenarios.

The papers discussion has been expanded to account for different possibilities which will help the readers get a more complete picture. I appreciate the review efforts and detailed response to reviewer comments.

<https://doi.org/10.7554/eLife.98358.2.sa2>

Reviewer #2 (Public review):

Summary:

Hiramatsu et al. investigated how cognate neurotransmitter receptors with antagonizing downstream effects localize within neurons when co-expressed. They focus on mapping the dopaminergic Dop1R1 and Dop2R receptors, corresponding to the mammalian D1- and D2-like dopamine receptors, which have opposing effects on intracellular cAMP levels, in neurons of the *Drosophila* mushroom body (MB). To visualize specific receptors in single neuron types within the crowded MB neuropil, the authors use existing dopamine receptor alleles tagged with 7 copies of split GFP to target the reconstitution of GFP tags specifically in the neurons of interest, providing a readout of receptor localization.

The authors demonstrate that both Dop1R1 and Dop2R are enriched, to differing degrees, in the axonal compartments of Kenyon cells cholinergic presynaptic inputs and in different dopamine neurons (DANs) that project axons to the MB. Co-localization studies of dopamine

receptors with the presynaptic marker Brp suggest that Dop1R1, and to a greater extent Dop2R, localize near release sites. This pattern in DANs suggests Dop1R1 and Dop2R serve as dual-feedback autoreceptors. Finally, they provide evidence that the balance of Dop1R1 and Dop2R in the axons of two different DAN populations is differentially modulated by starvation, which plays a role in regulating appetitive behaviors.

In their revised manuscript, Hiramatsu et al. revisited the localization and functional integrity of Dop1R1 and Dop2R within the *Drosophila* mushroom body. This revision strengthens their claims with new high-resolution imaging data and additional behavioral assays, supporting the functional integrity of 7X split GFP-tagged receptors and their distinct localizations within neural circuits.

The revised manuscript by Hiramatsu et al. demonstrates substantial improvements in experimental design and data presentation, effectively addressing concerns raised during the initial review. The addition of advanced imaging techniques and behavioral data confirms the functionality of tagged receptors, while providing deeper insights into their spatial and functional dynamics within neural circuits modulating responses to environmental changes like starvation. This study makes an important contribution to neuroscience, enhancing our understanding of dopamine receptor distribution in circuits underlying learning and memory.

Strengths:

The authors use reconstitution of GFP fluorescence of split GFP tags integrated at the endogenous locus of dopamine receptors, providing a precise readout of receptor localization. This method preserves endogenous transcriptional and post-transcriptional regulation, a critical feature for protein localization studies.

The choice of the *Drosophila* mushroom body as a model system is excellent, as it is well-studied, its connectome is carefully reconstructed, and its role in behaviors and associative memory enables linking receptor localization patterns to circuit function and behavior. This approach allows the authors to demonstrate that antagonizing dopamine receptors can act as autoreceptors within the axonal compartments of MB-innervating DANs. Moreover, they show that starvation differentially modulates the balance of these receptors in distinct DANs, highlighting the role of this regulation in circuit function and behavior.

The incorporation of higher-resolution Airyscan microscopy and functional assays in the revision provide evidence that tagged receptors retain functionality and predominantly localize at presynaptic sites within Kenyon cells and DANs. These findings support the dual autoreceptor feedback model proposed.

Weaknesses:

While the revision significantly strengthens the manuscript, the absence of specific antibodies against these receptors remains a limitation. This is understandable given the challenges of generating antibodies against such proteins. However, the use of more direct validation methods, such as specific antibodies (if available), and employing higher-resolution techniques like expansion microscopy, could further validate and enhance the robustness of the findings.

<https://doi.org/10.7554/eLife.98358.2.sa1>

Author response:

The following is the authors' response to the original reviews.

eLife Assessment

This study uses state-of-the-art methods to label endogenous dopamine receptors in a subset of Drosophila mushroom body neuronal types. The authors report that DopR1 and Dop2R receptors, which have opposing effects in intracellular cAMP, are present in axons termini of Kenyon cells, as well as those of two classes of dopaminergic neurons that innervate the mushroom body indicative of autocrine modulation by dopaminergic neurons. Additional experiments showing opposing effects of starvation on DopR1 and Dop2R levels in mushroom body neurons are consistent with a role for dopamine receptor levels increasing the efficiency of learned food-odour associations in starved flies. Supported by solid data, this is a valuable contribution to the field.

We thank the editors for the assessment, but request to change “DopR2” to “Dop2R”. The dopamine receptors in Drosophila have confusing names, but what we characterized in this study are called Dop1R1 (according to the Flybase; aka DopR1, dDA1, Dumb) and Dop2R (ibid; aka Dd2R). DopR2 is the name of a different dopamine receptor.

Public Reviews:

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Summary:

This is an important and interesting study that uses the split-GFP approach. Localization of receptors and correlating them to function is important in understanding the circuit basis of behavior.

Strengths:

The split-GFP approach allows visualization of subcellular enrichment of dopamine receptors in the plasma membrane of GAL4-expressing neurons allowing for a high level of specificity.

The authors resolve the presynaptic localization of DopR1 and Dop2R, in "giant" Drosophila neurons differentiated from cytokinesis-arrested neuroblasts in culture as it is not clear in the lobes and calyx.

Starvation-induced opposite responses of dopamine receptor expression in the PPL1 and PAM DANs provide key insights into models of appetitive learning.

Starvation-induced increase in D2R allows for increased negative feedback that the authors test in D2R knockout flies where appetitive memory is diminished.

This dual autoreceptor system is an attractive model for how amplitude and kinetics of dopamine release can be fine-tuned and controlled depending on the cellular function and this paper presents a good methodology to do it and a good system where the dynamics of dopamine release can be tested at the level of behavior.

Weaknesses:

LI measurements of Kenyon cells and lobes indicate that Dop2R was approximately twice as enriched in the lobe as the average density across the whole neuron, while the lobe enrichment of Dop1R1 was about 1.5 times the average, are these levels consistent during different times of the day and the state of the animal. How were these conditions controlled and how sensitive are receptor expression to the time of day of dissection, staining, etc.

To answer this question, we repeated the experiment in two replicates at different times of day and confirmed that the receptor localization was consistent (Figure 3 – figure supplement 1); LI measurements showed that Dop2R is enriched more in the lobe and less in the calyx compared to Dop1R1 (Figure 3D). The states of animals that could affect LI (e.g. feeding state and anesthesia for sorting, see methods) were kept constant.

The authors assume without discussion as to why and how presynaptic enrichment of these receptors is similar in giant neurons and MB.

In the revision, we added a short summary to recapitulate that the giant neurons exhibit many characteristics of mature neurons (Lines #152-156): "Importantly, these giant neurons exhibit characteristics of mature neurons, including firing patterns (Wu et al., 1990; Yao & Wu, 2001; Zhao & Wu, 1997) and acetylcholine release (Yao et al., 2000), both of which are regulated by cAMP and CaMKII signaling (Yao et al., 2000; Yao & Wu, 2001; Zhao & Wu, 1997)." In addition, we found punctate Brp accumulations localized to the axon terminals of the giant neurons (former Figure 4D and 4E). Therefore, the giant neuron serves as an excellent model to study the presynaptic localization of dopamine receptors in isolated large cells.

Figures 1-3 show the expensive expression of receptors in alpha and beta lobes while Figure 5 focusses on PAM and localization in γ and β' projections of PAM leading to the conclusion that presynaptic dopamine neurons express these and have feedback regulation. Consistency between lobes or discussion of these differences is important to consider.

In the revised manuscript, we show data in the γ KCs (Figure 4C, Figure 5 - figure supplement 1) in addition to α/β KCs, and demonstrate the consistent synaptic localization of Dop1R1 and Dop2R as in α/β KCs (Figure 4B and 5A).

Receptor expression in any learning-related MBONs is not discussed, and it would be intriguing as how receptors are organized in those cells. Given that these PAMs input to both KCs and MBONs these will have to work in some coordination.

The subcellular localization of dopamine receptors in MBONs indeed provides important insights into the site of dopaminergic signaling in these neurons (Takemura et al., 2017; Pavlovsky et al., 2018; Pribbenow et al., 2022). Therefore, we added new data for Dop1R1 and Dop2R in MBON- γ 1pedc> $\alpha\beta$ (Figure 6). Interestingly, these receptors are localized to in the dendritic projection in the γ 1 compartment as well as presynaptic boutons (Figure 6).

Although authors use the D2R enhancement post starvation to show that knocking down receptors eliminated appetitive memory, the knocking out is affecting multiple neurons within this circuit including PAMs and KCs. How does that account for the observed effect? Are those not important for appetitive learning?

In the appetitive memory experiment (Figure 9C), we knocked down Dop2R only in the select neurons of the PPL1 cluster, and this manipulation does not directly affect Dop2R expression in PAMs and KCs.

Starvation-induced enhancement of Dop2R expression in the PPL1 neurons (Figure 8F) would attenuate their outputs and therefore disinhibit expression of appetitive memory in starved flies (Krashes et al., 2009). Consistently, Dop2R knock-down in PPL1 impaired appetitive memory in starved flies (Figure 9C). We revised the corresponding text to make this point clearer (Lines #224227).

The evidence for fine-tuning is completely based on receptor expression and one behavioral outcome which could result from many possibilities. It is not clear if this fine-tuning and presynaptic feedback regulation-based dopamine release is a clear possibility. Alternate hypotheses and outcomes could be considered in the model as it is not completely substantiated by data at least as presented.

The reviewer's concern is valid, and the presynaptic dopamine tuning by autoreceptors may need more experimental support. We therefore additionally discussed another possibility (Lines #289-291): "Alternatively, these presynaptic receptors could potentially receive extrasynaptic dopamine released from other DANs. Therefore, the autoreceptor functions need to be experimentally clarified by manipulating the receptor expression in DANs."

Reviewer #2 (Public Review):

Summary:

Hiramatsu et al. investigated how cognate neurotransmitter receptors with antagonizing downstream effects localize within neurons when co-expressed. They focus on mapping the localization of the dopaminergic Dop1R1 and Dop2R receptors, which correspond to the mammalian D1- and D2-like dopamine receptors, which have opposing effects on intracellular cAMP levels, in neurons of the Drosophila mushroom body (MB). To visualize specific receptors in single neuron types within the crowded MB neuropil, the authors use existing dopamine receptor alleles tagged with 7 copies of split GFP to target reconstitution of GFP tags only in the neurons of interest as a read-out of receptor localization. The authors show that both Dop1R1 and Dop2R, with differing degrees, are enriched in axonal compartments of both the Kenyon Cells cholinergic presynaptic inputs and in different dopamine neurons (DANs), which project axons to the MB. Co-localization studies of dopamine receptors with the presynaptic marker Brp suggest that Dop1R1 and, to a larger extent Dop2R, localize in the proximity of release sites. This localization pattern in DANs suggests that Dop1R1 and Dop2R work in dual-feedback regulation as autoreceptors. Finally, they provide evidence that the balance of Dop1R1 and Dop2R in the axons of two different DAN populations is differentially modulated by starvation and that this regulation plays a role in regulating appetitive behaviors.

Strengths:

The authors use reconstitution of GFP fluorescence of split GFP tags knocked into the endogenous locus at the C-terminus of the dopamine receptors as a readout of dopamine receptor localization. This elegant approach preserves the endogenous transcriptional and post-transcriptional regulation of the receptor, which is essential for studies of protein localization.

The study focuses on mapping the localization of dopamine receptors in neurons of the mushroom body. This is an excellent choice of system to address the question posed in this study, as the neurons are well-studied, and their connections are carefully reconstructed in the mushroom body connectome. Furthermore, the role of this circuit in different behaviors and associative memory permits the linking of patterns of receptor localization to circuit function and resulting behavior. Because of these features, the authors can provide evidence that two antagonizing dopamine receptors can act as autoreceptors within the axonal compartment of MB innervating DANs. The differential regulation of the balance of the two receptors under starvation in two distinct DAN innervations provides evidence of the role that regulation of this balance can play in circuit function and behavioral output.

Weaknesses:

The approach of using endogenously tagged alleles to study localization is a strength of this study, but the authors do not provide sufficient evidence that the insertion of 7 copies of split GFP to the C terminus of the dopamine receptors does not interfere with the endogenous localization pattern or function. Both sets of tagged alleles (1X Venus and 7X split GFP tagged) were previously reported (Kondo et al., 2020), but only the 1X Venus tagged alleles were further functionally validated in assays of olfactory appetitive memory. Despite the smaller size of the 7X split-GFP array tag knocked into the same location as the 1X venus tag, the reconstitution of 7 copies of GFP at the C terminus of the dopamine receptor, might substantially increase the molecular bulk at this site, potentially impeding the function of the receptor more significantly than the smaller, single Venus tag. The data presented by Kondo et al. 2020, is insufficient to conclude that the two alleles are equivalent.

In the revision, we validated the function of these engineered receptors by a new set of olfactory learning experiments. Both these receptors in KCs were shown to be required for aversive memory (Kim et al., 2007, Scholz-Kornehl et al., 2016). As in the anatomical experiments, we induced GFP110 expression in KC of the flies homozygous for 7xGFP₁₁-tagged receptors using *MB-Switch* and 3 days of RU486 feeding o. We confirmed STM performance of these flies were not significantly different from the control (Figure 2 – figure supplement 1). Thus, these fusion receptors are functional.

The authors' conclusion that the receptors localize to presynaptic sites is weak. The analysis of the colocalization of the active zone marker Brp whole-brain staining with dopamine receptors labeled in specific neurons is insufficient to conclude that the receptors are localized at presynaptic sites. Given the highly crowded neuropil environment, the data cannot differentiate between the receptor localization postsynaptic to a dopamine release site or at a presynaptic site within the same neuron. The known distribution of presynaptic sites within the neurons analyzed in the study provides evidence that the receptors are enriched in axonal compartments, but co-labeling of presynaptic sites and receptors in the same neuron or super-resolution methods are needed to provide evidence of receptor localization at active zones. The data presented in Figures 5K-5L provides compelling evidence that the receptors localize to neuronal varicosities in DANs where the receptors could play a role as autoreceptors.

Given the highly crowded environment of the mushroom body neuropil, the analysis of dopamine receptor localization in Kenyon cells is not conclusive. The data is sufficient to conclude that the receptors are preferentially localizing to the axonal compartment of Kenyon cells, but co-localization with brain-wide Brp active zone immunostaining is not sufficient to determine if the receptor localizes juxtaposed to dopaminergic release sites, in proximity of release sites in Kenyon cells, or both.

To better resolve the microcircuits of KCs, we triple-labeled the plasma membrane and DAR::rGFP in KCs, and Brp, and examined their localizations with high-resolution imaging with Airyscan. This strategy revealed the receptor clusters associated with Brp accumulation within KCs (Figure 4). To further verify the association of DARs and active zones within KCs, we co-expressed Brp^{short}::mStraw and GFP₁₋₁₀ and confirmed their colocalization (Figure 5A), suggesting presynaptic localization of DARs in KCs. With these additional characterizations, we now discuss the significance of receptors at the presynaptic sites of KCs.

Recommendations for the authors:

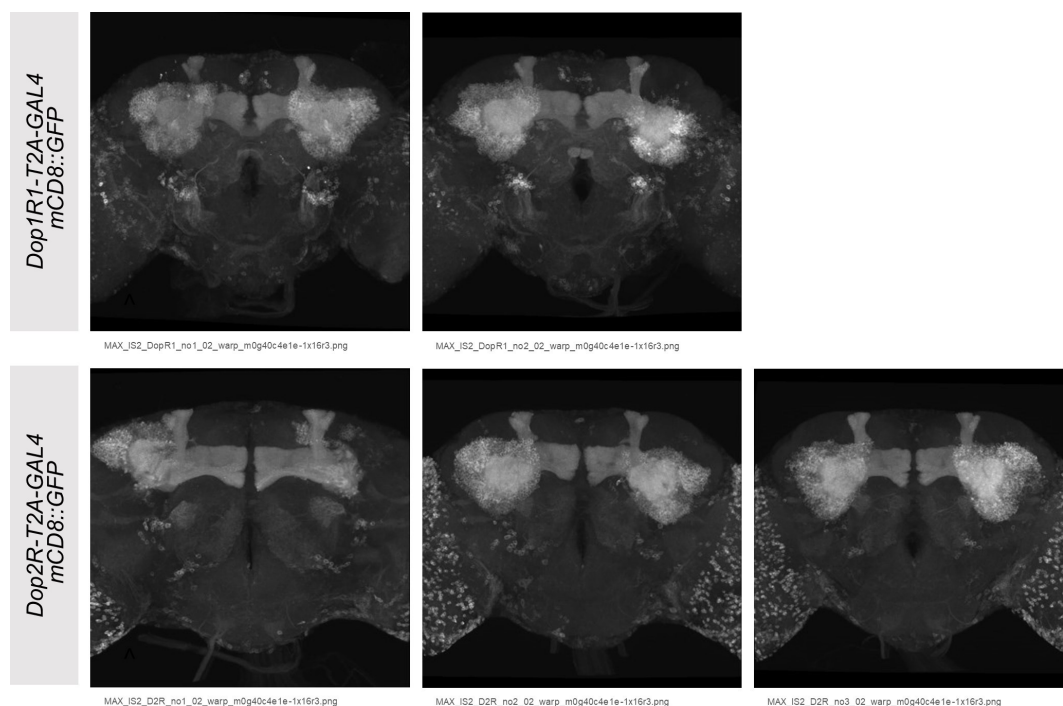
Reviewer #1 (Recommendations For The Authors):

This is an important and interesting study that uses the split-GFP approach. Localization of receptors and correlating them to function is important in understanding the circuit basis of behavior.

For Figure 1, the authors show PAM, PPL1 neurons, and the ellipsoid body as a validation of their tools (Dop1R1-T2A-GAL4 and Dop2R-T2A-GAL4) and the idea that these receptors are colocalized. However, it appears that the technique was applied to the whole brain so it would be great to see the whole brain to understand how much labelling is specific and how stochastic. Methods could include how dissection conditions were controlled and how sensitive are receptor expression to the time of day of dissection, staining, etc.

The expression patterns of the receptor T2A-GAL4 lines (Figure 1A and 1B) are consistent in the multiple whole brains (Kondo et al., 2020, Author response image 1).

Author response image 1.



The significance of the expression of these two receptors in an active zone is not clearly discussed and presynaptic localization is not elaborated on. Would something like expansion microscopy be useful in resolving this? It would be important to discuss that as giant neurons in culture don't replicate many aspects of the MB system.

In the revised manuscript, we elaborated discussion regarding the function of the two antagonizing receptors at the AZ (Lines #226-275).

Does MB-GeneSwitch > GFP1-1 reliably express in gamma lobes? Most of the figures show alpha/beta lobes.

Yes. MB-GeneSwitch is also expressed in γ KCs, but weakly. 12 hours of RU486 feeding, which we did in the previous experiments, was insufficient to induce GFP reconstitution in the γ

KCs. By extending the time of transgene induction, we visualized expression of Dop1R1 and Dop2R more clearly in γ KCs. Their localization is similar to that in the α/β KCs (Figure 4C, Figure 5 - figure supplement 1).

Figure 6, y-axis says protein level. At first, I thought it was related to starvation so maybe authors can be more specific as the protein level doesn't indicate any aspect of starvation.

We appreciate this comment, and the labels on the y-axis were now changed to “rGFP levels” (Figure 8C and 8F, Figure 8 - figure supplement 1B, 1D and 1F).

Reviewer #2 (Recommendations For The Authors):

Title:

The title of the manuscript focuses on the tagging of the receptors and their synaptic enrichment.

Given that the alleles used in the study were generated in a previously published study (Kondo et al, 2020), which describes the receptor tagging and that the data currently provided is insufficient to conclude that the receptors are localizing to synapses, the title should be changed to reflect the focus on localizing antagonistic cognate neurotransmitter receptors in the same neuron and their putative role as autoreceptors in DANs.

Following this advice, we removed the methodology from the title and revised it to “Synaptic enrichment and dynamic regulation of the two opposing dopamine receptors within the same neurons”.

Minor issues with text and figures:

Figure 1

A conclusion from Figure 1 is that the two receptors are co-expressed in Kenyon cells. Please provide panels equivalent to the ones shown in D-G, with Kenyon cells cell bodies, or mark these cells in the existing panels, if present. Line 111 refers to panel 1D as the Kenyon cells panel, which is currently a PAM panel.

We added images for coexpression of these receptors in the cell bodies of KCs (Figure 1 - figure supplement 1) and revised the text accordingly (Lines #89-90).

Given that most of the study centers on visualizing receptor localization, it would benefit the reader to include labels in Figure 1 that help understand that these panels reflect expression patterns rather than receptor localization. For instance, rCD2::GFP could be indicated in the Dop1R1-LexA panels.

As suggested, labels were added to indicate the UAS and lexAop markers (Figure 1D, 1E, 1G-1I and Figure 1 – figure supplement 1).

Given that panels D-E focus on the cell bodies of the neurons, it could be beneficial for the reader to present the ellipsoid body neurons using a similar view that only shows the cell bodies. Similarly, one could just show the glial cell bodies .

We now show the cell bodies of ring neurons (Figure 1G) and ensheathing glia (Figure 1I).

For panel 1E, please indicate the subset of PPL1 neurons that both expressed Dop1R1 and Dop2R, as indicated in the text, as it is currently unclear from the image.

Dop1R1-T2A-LexA was barely detected in all PPL1 (Figure 1E). We corrected the confusing text (Lines #95-96).

Figure 2

The cartoon of the cell-type-specific labeling should show that the tag is 7XFP-11 and the UAScomponent FP-10, as the current cartoon leads the reader to conclude that the receptors are tagged with a single copy of split GFP. The detail that the receptors are tagged with 7 copies of split GFP is only provided through the genotype of the allele in the resource table. This design aspect should be made clear in the figure and the text when describing the allele and approach used to tag receptors in specific neuron types.

We now added the construct design in the scheme (Figure 2A) and revised the corresponding text (Line #101-103).

Panel A. The arrow representing the endogenous promoter in the yellow gene representation should be placed at the beginning of the coding sequence. Currently, the different colors of what I assume are coding (yellow) and non-coding (white) transcript regions are not described in the legend. I would omit these or represent them in the same color as thinner boxes if the authors want to emphasize that the tag is inserted at the C terminus within the endogenous locus.

The color scheme was revised to be more consistent and intuitive (Figure 2A).

Figure 3

Labels of the calyx and MB lobes would benefit readers not as familiar with the system used in the study. In addition, it would be beneficial to the reader to indicate in panel A the location of the compartments analyzed in panel H (e.g., peduncle, a3).

Figure 3A was amended to clearly indicate the analyzed MB compartments.

Adding frontal and sagittal to panels B-E, as in Figure 2, would help the reader interpret the data.

In Figure 3B, “Frontal” and “Sagittal” were indicated.

Panel F-G. A scale bar should be provided for the data shown in the insets. Could the author comment on the localization of Dop1R1 in KCs? The data in the current panel suggests that only a subset of KCs express high levels of receptors in their axons, as a portion of the membrane is devoid of receptor signals. This would be in line with differential dopamine receptor expression in subsets of Kenyon cells, as shown in Kondo et al., 2020, which is currently not commented on in the paper.

We confirmed that the majority of the KCs express both *Dop1R1* and *Dop2R* genes (Figure 1 - figure supplement 1). LIs should be compared within the same cells rather than the differences of protein levels between cell types as they also reflect the GAL4 expression levels.

Panel H. Some P values are shown as n.s. ($p > 0.05$). Other non-significant p values in this panel and in other figures throughout the paper are instead reported (e.g. peduncle

P=0.164). For consistency, please report the values as n.s. as indicated in the methods for all non-significant tests in this panel and throughout the manuscript.

We now present the new dataset, and the graph represents the appropriate statistical results (Figure 3D; see the methods section for details).

The methods of labeling the receptors through the expression of the GeneSwitch-controlled GFP1-10 in Kenyon cells induced by RU486 are not provided in the methods. Please provide a description of this as referenced in the figure legend and the genotypes used in the analysis shown in the panels.

The method of RU486 feeding has been added. We apologize for the missing method.

Figure 4

Please provide scale bars for the inset in panels A-B.

Scale bars were added to all confocal images.

The current analysis cannot distinguish between postsynaptic and presynaptic dopamine receptors in KCs, and the figure title should reflect this.

We now present the new data dopamine receptors in KCs and clearly distinguish Brp clusters of the KCs and other cell types (Figure 4, Figure 5).

The reader could benefit from additional details of using the giant neuron model, as it is not commonly used, and it is not clear how to relate this to interpret the localization of dopaminergic receptors within Kenyon cells. The use of the venus-tagged receptor variant should be introduced in the text, as using a different allele currently lacks context. Figures 4F-4J show that the receptor is localizing throughout the neuron. Quantifying the fraction of receptor signal colocalizing with Brp could aid in interpreting the data. However, it would still not be clear how to interpret this data in the context of understanding the localization of the receptors in neurons within fly brain circuits. In the absence of additional data, the data provided in Figure 4 is inconclusive and could be omitted, keeping the focus of the study on the analysis of the two receptors in DANs. Co-expressing a presynaptic marker in Kenyon cells (e.g., by expressing Brp::SNAP) in conjunction with rGFP labeled receptor would provide additional evidence of the relationship of release sites in Kenyon cells and tagged dopamine receptors in these same cells and could add evidence in support to the current conclusion.

Following the advice, we added a short summary to recapitulate that the giant neurons exhibit many characteristics of mature neurons (Lines #152-156): "Importantly, these giant neurons exhibit characteristics of mature neurons, including firing patterns (Wu et al., 1990; Yao & Wu, 2001; Zhao & Wu, 1997) and acetylcholine release (Yao et al., 2000), both of which are regulated by cAMP and CaMKII signaling (Yao et al., 2000; Yao & Wu, 2001; Zhao & Wu, 1997)." Therefore, the giant neuron serves as an excellent model to study the presynaptic localization in large cells in isolation.

To clarify polarized localization of Brp clusters and dopamine receptors but not "localizing throughout the neuron", we now show less magnified data (Figure 5C). It clearly demonstrates punctate Brp accumulations localized to the axon terminals of the giant neurons (former Figure 4D and 4E). This is the same membrane segment where Dop1R1 and Dop2R are localized (Figure 5C). Therefore, the association of Brp clusters and the dopamine receptors in the isolated giant neurons suggests that the subcellular localization in the brain neurons is independent of the circuit context.

As the giant neurons do not form intermingled circuits, venus-tagged receptors are sufficient for this experiment and simpler in genetics.

Following the suggestion to clarify the AZ association of the receptors in KCs, we coexpressed Brpshort-mStraw and GFP1-10 in KCs and confirmed their colocalization (Figure 5A).

Figure 6

The data and analysis show that starvation induces changes in the $\alpha 3$ compartment in PPL1 neurons only, while the data provided shows no significant change for PPL1 neurons innervating other MB compartments. This should be clearly stated in lines 174-175, as it is implied that there is a difference in the analysis for compartments other than $\alpha 3$. Panel L of Figure 6 - supplement 1 shows no significant change for all three compartments analyzed and should be indicated as n.s. in all instances, as stated in the methods.

We revised the text to clarify that the starvation-induced differences of Dop2R expression were not significant (Lines #217-219). The reason to highlight the $\alpha 3$ compartment is that both Dop1R1 and Dop2R are coexpressed in this PPL1 neuron (Figure 8D).

Additional minor comments:

There are a few typos and errors throughout the manuscript. The text should be carefully proofread to correct these. Here are the ones that came to my attention:

Please reference all figure panels in the text. For instance, Figure 3A is not mentioned and should be revised in line 112 as Figure 3A-E.

Lines 103-104. The sentence "LI was visualized as the color of the membrane signals" is unclear and should be revised.

Figure 4 legend - dendritic claws should likely be B and C and not B and E.

Lines 147 - Incorrect figure panels, should be 5C-L or 5D-E.

Line 241 - DNAs should be DANs.

Methods - please define what the abbreviation CS stands for.

We really appreciate for careful reading of this reviewer. All these were corrected.

<https://doi.org/10.7554/eLife.98358.2.sa0>